D-SIMM: deterministic margin limits for decentralised lending

Alexander McFarlane

August 31, 2026

Abstract

Many tokenised assets, including funds, structured products, and corporate bonds, are valued only by their issuer, its administrator, or a primary dealer, so their value is a periodic published valuation rather than a price quoted every block. Decentralised lending's margining assumptions fit neither that valuation basis nor a liquidation that often takes weeks. Where a secondary market exists, liquidity in large amounts is rarely available within a block, and the only guaranteed exit is redemption with the issuer. D-SIMM applies the International Swaps and Derivatives Association's Standard Initial Margin Model (SIMM) to margining in decentralised finance (DeFi). Off-chain risk is what SIMM was built to margin. In this first iteration we focus on the most popular case, tokenised funds. We derive narrow bounds from published risk weights for loan-to-value (LTV) limits, liquidation thresholds (LLTV, liquidation loan-to-value), and the borrow and supply caps. A liquidated position is assumed to be held through the clearing window to cash settlement. Free parameters remain for the practitioner to choose, but the aim is to reduce the impact of discretion on leverage in DeFi as far as possible, by setting transparent limits.

Acknowledgements: The author thanks Sahil Gujrati for proofreading earlier drafts and identifying numerous errors, and Sebastien Guglietta for useful discussions.

1 Overview

Margining collateral that has no continuous market price is a problem traditional finance has already solved for bilateral derivatives and trading-book capital, in two components. The International Swaps and Derivatives Association's Standard Initial Margin Model (ISDA SIMM) prescribes sensitivity-based initial margin from published risk weights and correlations [1]. The Basel Fundamental Review of the Trading Book (FRTB) market-risk standard prescribes liquidity horizons and conservative floors where market data does not exist [2]. D-SIMM combines the two into limits for lending markets, and this paper develops it for tokenised funds, whose shares exit through issuer redemption at the published valuation, the per-share valuation the issuer or its administrator publishes on a stated frequency.

This paper specifies the model. The public companion *Applying D-SIMM* contains the data, parameter-assignment, validation, and implementation material.

A lending market has two sides. Suppliers deposit a tokenised asset as collateral, and borrowers draw cash against it. The over-collateralisation the market demands, one minus the loan-to-value ratio, plays the role initial margin plays on a derivative: it reserves against the move in collateral value before the position can be closed. When a position crosses its liquidation threshold, the deployment, the lending market instance holding the collateral, recovers its cash through the issuer's redemption process. Each of the four outputs governs one part of the risk of that recovery, for each collateral, a.

The borrow limit $\mathrm{LTV}^{(a)}$ is the loan-to-value ratio at which no further borrowing is permitted. The liquidation threshold $\mathrm{LLTV}^{(a)}$ (liquidation loan-to-value) is the ratio at which a position becomes eligible for liquidation. The liquidation buffer between the two gives a borrower room before recovery starts. The borrow cap $B^{(a)}_{\max}$ bounds the debt outstanding against the collateral, so that one dealing window clears it. The supply cap $S^{(a)}_{\max}$ bounds the collateral the market accepts, so that the whole portfolio clears within a small number of dealing windows. In what follows we write $\theta^{(a)} := \mathrm{LLTV}^{(a)}$ and $\ell^{(a)}_{\max} := \mathrm{LTV}^{(a)}$, so that ratios such as $\ell^{(a)}_{\max}/\theta^{(a)}$ express proximity to liquidation in normalised units. Figure 1 outlines the framework: attested terms and published risk weights feed the D-SIMM haircut engine, and the haircut sets the on-chain limits. Due diligence admits or rejects an asset, and monitoring can only reduce the limits for new exposure.

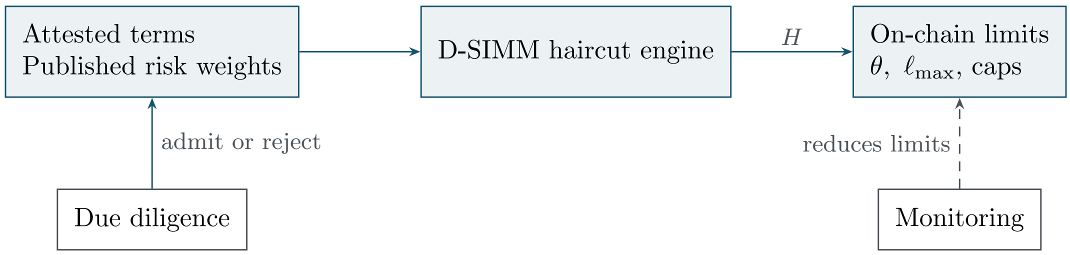

Figure 1: Attested terms and published risk weights feed the D-SIMM haircut engine. The haircut H sets the active on-chain limits with no iteration. Due diligence admits or rejects an asset, and monitoring can only reduce those limits.

1.1 Universe and settlement

A tokenised asset is a token whose ownership is governed by existing legal frameworks. Some have physical backing held off-chain, such as gold and share certificates. Assets without physical backing may in time be governed entirely by on-chain ownership rights, derivatives and equities among them. Few such assets exist today. The collateral universe is tokenised assets that correspond to an underlying fund. The risk factors every type is charged against are fixed in Section 3, and the practitioner defines their own types against them.

Crypto-native assets are out of scope. A crypto-native asset is a natively issued crypto asset, Bitcoin and Ether among them, that represents no off-chain asset or liability, whether directly or through a derivative. The definition separates a tokenised dollar and natively issued company stock from the crypto-native class. Many assets commonly described as "crypto-native", a fiat-backed stablecoin among them, have off-chain assets and liabilities and so meet the definition of a tokenised asset. Whether the framework evaluates one still turns on the fund correspondence above.

The fund share has the exit the framework charges for: redemption with the issuer at the published valuation over a contractual window. An extension to non-fund assets would rest on identifying how each asset settles. Settlement is either with an issuer or primary dealer under contracted redemption or trading terms, or in a market the practitioner can show to be deep from public data. Justifying liquidity without public data would take a strong evidence standard.

Bridge risk sits outside the scope of this framework. An asset whose token, administration, or valuation oracle depends on a bridge or cross-chain messaging system is not evaluated under the framework.

Feeder structures are eligible. A feeder token holds its economic exposure through a master fund. The look-through of Section 3.2.4 extends to the master's holdings, read from the master's own documents, so the sensitivities the margin is computed from are the master's. Three master-level risks take named layers (Section 2.3). Gate pass-through takes the gating layer, and master valuation staleness takes the valuation-staleness layer. The additional administrator in the chain takes the manager/operational layer. The practitioner sets the evidence standard the master-level legs are held to (free parameter, Appendix A).

Most tokenised assets trade too thinly to offer instant liquidity in large amounts, and that is the ordinary condition of the assets themselves. Their off-chain equivalents settle over days or weeks, and traditional clearing is built around periods of time rather than instant price discovery. *The framework assumes subscription and redemption at the published valuation only.*[1] Any automated-market-maker (AMM), on-chain secondary market, slippage, maximal extractable value (MEV), gas, or atomic-withdrawal route is out of scope and deferred to a later draft.

The cost of clearing a position in stress rises with its size, and the framework charges for it. SIMM's risk weights are calibrated on stressed histories. The concentration risk factor steepens the charge once a position outgrows one dealing window of attested capacity. The clearing discount rescales the stressed move to the whole clearing window. What redemption at the published valuation removes is the modelling of venue depth.

The primary object is the liquidity horizon and the process that settles it, which for a tokenised fund is the issuer's own subscription and redemption. The horizon runs from the breach to cash: the worst-case wait to the next submission deadline plus the attested notice, dealing, and settlement legs (Eq. (4)). Other tokenised assets have other settlement processes and other horizons, and Section 2.1 sets out the three sources those legs are read from, in a fixed order. Redemption at the published valuation is the settlement the framework treats as guaranteed, in stress as well as in normal conditions.

Holding-period risk remains, and it falls on both sides. A borrower waiting to release collateral and the deployment recovering a position in liquidation wait for the same window, and neither can shorten it. The clearing discount $\delta^{(a)}_{\text{clear}}$ reserves against that wait. Section 2.2 sets it precisely. The framework sets the discount per asset, and stays silent on the clearing mechanism and on how the discount is enforced or paid.

1.2 Outputs and how they are set

All four outputs are computed in a single pass from attested terms and published risk weights. The interest cost depends on the liquidation threshold, and the two resolve together in closed form, so there is no circular dependency (Eq. (8)). The haircut $H^{(a)}$ is the fraction of the published valuation reserved against holding a position in liquidation to cash settlement, built as a four-part sum. The liquidation threshold $\theta^{(a)}$ is what remains of the published valuation after the haircut, scaled by the due-diligence tier multiplier $\kappa^{(a)} \leq 1$ ($\kappa^{(a)} = 1$ at Prime; Section 5.2). The borrow limit $\ell^{(a)}_{\max}$ sits the liquidation buffer $\nu^{(a)}$, 4–8 percentage points, below $\theta^{(a)}$. The buffer is set per listing (Section 5.1). Both caps are keyed to the issuer's attested redemption capacity $R^{(a)}_{\text{cap}}$. The borrow cap is set so that one dealing window clears the pool's entire debt at the point of liquidation, and the supply cap is an elected multiple M of that same capacity:

$$H^{(a)} = \mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \delta^{(a)}_{\text{carry}} + \Lambda^{(a)}, \qquad \theta^{(a)} = \kappa^{(a)}\big(1 - H^{(a)}\big), \qquad \ell^{(a)}_{\max} = \theta^{(a)} - \nu^{(a)},$$
$$B^{(a)}_{\max} \;\leq\; R^{(a)}_{\text{cap}}, \qquad S^{(a)}_{\max} \;\leq\; M\,R^{(a)}_{\text{cap}}.$$

The borrow cap is further bounded by the committed settlement level, read from the subscription documents case by case as the default level. Where the documents commit to no measurable level the borrow cap is zero, and the asset is listed and not borrowed against (Section 5.4).

The practitioner sets the model constants outside the calculation. The supply-cap multiple M is an open model quantity (free parameter, Appendix A; $M = 8$ is illustrative). Section 2 states the recipe in full and defines each symbol (Eq. (1)). The four haircut components are:

(i) $\mathrm{IM}^{(a)}$: the delta component of the ISDA SIMM v2.6 initial margin per unit of published valuation [1], taken verbatim at SIMM's native ten-business-day horizon. The rescaling to the governing clearing window happens in $\delta^{(a)}_{\text{clear}}$.

[1] Transfer restrictions can require any party taking delivery of the collateral, a liquidator included, to hold know-your-customer approval from the issuer.

(ii) $\delta^{(a)}_{\text{clear}}$: the *clearing discount*, the difference between the price-risk charge over the clearing window $\text{LH}^{(a)}$ and the margin over ten business days, set by Eq. (6). It is a charge where the window runs beyond ten business days, and a credit where contractually enforceable terms, attested or documented, enforce a shorter one. Estimated legs and FRTB fallback windows never earn one. Qualified clearing observations can take the total price-risk charge above the rescaled move, which floors it.

(iii) $\delta^{(a)}_{\text{carry}}$: the interest cost of holding the debt through the clearing window, resolved in closed form at Eq. (8).

(iv) $\Lambda^{(a)}$: the sum of additive layers (gating/suspension, valuation staleness/smoothing, default add-on, oracle pipeline integrity, prepayment, manager/operational, wrong-way, leverage add-on, event risk), charging for the risks SIMM structurally cannot see. Each layer is defined in Section 2.3. The practitioner sets the model constants used to determine the layer values.

Concentration enters inside $\text{IM}^{(a)}$ through the multiplier $\text{CR} = \max\big(1, \sqrt{\text{position}/T}\big)$, with the threshold T re-keyed to the attested issuer redemption capacity $R^{(a)}_{\text{cap}}$. $R^{(a)}_{\text{cap}}$ is attested per window, window-normalised where the dealing window differs from the basis on which the capacity is attested (Section 2.5). It is attested against the contractual dealing terms only, so a holding inside the asset that could fund redemption faster than the contract requires counts as zero capacity unless the issuer attests a committed fast-redemption capacity. A committed fast-redemption component enters $R^{(a)}_{\text{cap}}$ only when the issuer attests it.

Any unavailable input takes a prescribed conservative floor, never zero (Section 4.6). Controlled records hold the recomputation schedule. The deployed market configuration is the public source for each active LTV, LLTV, borrow cap, and supply cap. Section 4.8 works a treasury money-market fund ($H = 2.23\%$) end to end.

The practitioner's due-diligence scorecard governs eligibility and tiering. An eligibility failure rejects the asset outright. Otherwise the assigned tier scales the over-collateralisation cushion by the tier multiplier $\kappa^{(a)} \leq 1$ (illustratively $\{1.00, 0.90, 0.75\}$ for a Prime/Standard/Restricted structure), and it acts on that cushion alone (Section 5.2). The multiplier is set at listing from due-diligence quality and re-scored at the governed frequency. An adverse re-score is applied through the new-borrow limit and the caps, never onto the liquidation threshold of a live position.

1.3 Emergency actions

All emergency authority acts on *new* exposure and capacity, never on the liquidation threshold of live positions. Lowering the liquidation threshold on live collateral exactly when the exit route is slowest, on a gated or suspended asset, is procyclical and self-defeating: it would create the forced-selling spiral this design removes. Five actions are available, in increasing severity. The mildest is the new-borrow LTV cut, a lower $\ell^{(a)}_{\max}$ for new borrows only with live positions keeping their limit. Above it sit the cap cut, which reduces $B^{(a)}_{\max}$ and $S^{(a)}_{\max}$, and the origination rate limit, which slows new borrows per dealing window. The two severest are redemption-only mode, in which no new borrows open and collateral exits only through issuer redemption, and the pause, under which the collateral accepts no new borrows and no new supply. The escalation ladder implementing these actions is set out in Section 5.6.

Guardrails are valuation-native and observable from day one. The operations companion lists these observables, including rating events. Section 5.6 states the rules they implement. Among the rating events, loss of investment grade on rated securitised collateral forces escalation level 3, and *confirmed* loss forces automatic full redemption at escalation level 4. Full redemption files redemption notices at contractual dealing dates with no acceleration, an orderly exit of the asset that still never lowers the liquidation threshold on a live position (Section 5.6).

The deployed market configuration shows one liquidation threshold per market. An adverse

$\theta^{(a)}$ change acts through new-borrow $\ell^{(a)}_{\max}$, caps, and pauses. The live liquidation threshold stays where it was set.

2 Theoretical framework

The collateral universe is the tokenised funds of Section 1.1, whose only exit is issuer redemption at the published valuation over contractual windows. The exit is a process rather than a trade. The framework therefore models no venue depth, and reserves the cost of exiting in stress through the stressed SIMM risk weights, the concentration risk factor, and the clearing discount instead. Any incidental on-chain liquidity counts as zero, out of prudence. The theory reduces to three objects: a price process that jumps only at valuation publications, a worst-case loss over the contractual exit horizon, and a set of closed-form limits derived from them.

Per collateral a, the recipe is

$$H^{(a)} = \mathrm{IM}^{(a)} + \delta^{(a)}_{\mathrm{clear}} + \delta^{(a)}_{\mathrm{carry}} + \Lambda^{(a)}, \qquad \theta^{(a)} = \kappa^{(a)}\big(1 - H^{(a)}\big), \qquad \ell^{(a)}_{\max} = \theta^{(a)} - \nu^{(a)}, \tag{1}$$

(the tier multiplier $\kappa^{(a)} \le 1$, a free parameter of the practitioner's tier structure, scales the range; $\kappa = 1$ at the Prime tier, so $\theta = 1 - H$ there) with caps $B^{(a)}_{\max} \le R^{(a)}_{\mathrm{cap}}$ and $S^{(a)}_{\max} \le M\, R^{(a)}_{\mathrm{cap}}$ (Section 2.5). Here $\mathrm{IM}^{(a)}$ is the SIMM delta margin of Eq. (5), $\delta^{(a)}_{\mathrm{clear}}$ the clearing discount of Eq. (6), $\delta^{(a)}_{\mathrm{carry}}$ the interest cost of Eq. (7), $\Lambda^{(a)}$ the layer total of Eq. (9), and $\nu^{(a)}$ the per-listing liquidation buffer of Section 5.1. Every input is a published SIMM parameter, an attested redemption term (the issuer- or transfer-agent-attested notice, dealing-window, and settlement-lag figures of Section 2.1), an observed clearing cost, or a fixed policy constant. Nothing is fitted to market data that does not exist for this universe.

2.1 Setting and notation

Fix a collateral a. The official published valuation per token $N^{(a)}_k$ is published by the administrator at publication times $t_0 < t_1 < \cdots$ with publication cycle $\Delta^{(a)} = t_{k+1} - t_k$ set by the attested publication calendar: daily for a typical money-market fund (MMF), up to monthly for private credit and hedge funds. Between publications the protocol price accrues deterministically at the attested yield $y^{(a)}$, the portfolio yield net of fees, attested by the administrator alongside the valuation feed:

$$P^{(a)}_t = \tilde{N}^{(a)}_t = N^{(a)}_k\big(1 + y^{(a)}(t - t_k)\big), \qquad t \in [t_k, t_{k+1}), \tag{2}$$

and all price risk realises as a jump at the next publication,

$$J^{(a)}_{k+1} = \frac{N^{(a)}_{k+1}}{\tilde{N}^{(a)}_{t^-_{k+1}}} - 1. \tag{3}$$

The assets in scope have no continuous secondary-market price: intraday variance, order-flow impact, and oracle-versus-market deviation are not defined objects for this process. A published valuation does not by itself imply redeemability: gates, suspensions, and side-pockets can narrow the exit while valuations continue to be published. That state has no object in this price process. It is charged by the gating/suspension layer of Section 2.3 and managed by the gate procedures of Section 5.6.

Exiting a position means redeeming with the issuer. A forced exit cannot time the submission calendar, so the clearing window runs from the breach to cash: the worst-case wait w to the next submission deadline, then the attested notice period d_n, dealing window d_d, and settlement lag d_s (all in business days),

$$\mathrm{LH}^{(a)} = \max\big\{1,\ w^{(a)} + d^{(a)}_n + d^{(a)}_d + d^{(a)}_s\big\}. \tag{4}$$

$\mathrm{LH}^{(a)}$ is a contractual, attestable quantity rather than an estimate. For an asset dealing on fixed calendar dates the wait varies with the breach date, and the maximum is taken. Averages are never used.

Three sources fix the legs, in a fixed order. A written service-level agreement granted by the issuer or transfer agent in the attested redemption terms fixes them where one exists. It replaces the regulatory horizon rather than taking the larger of the two, so a shorter window is a reduction earned by evidence: a securitisation-type collateral with an attested redemption service-level agreement of 45 business days takes LH = 45 business days plus its worst-case wait. Where no such agreement exists but the issuer's own offering documents state a redemption or liquidity horizon, the framework reads d_n, d_d, and d_s from those contractual statements, with the worst-case wait still added. Where neither exists, the regulatory liquidity horizon for the exposure under the Fundamental Review of the Trading Book (FRTB) [2] applies: 40 business days for investment-grade credit, 60 for high yield, and 120 for securitisation. That row is floored at the contractual dealing window wherever that window is longer, and the worst-case wait is added to it as well, so the same collateral with neither takes LH = 120 business days plus its worst-case wait, the securitisation floor.

Where a class has no natural FRTB horizon, as with event-linked collateral, the practitioner sets a conservative fallback window as a model constant. A gating provision in the offering documents is charged separately in the gating layer rather than lengthening the window, so the contractual horizon and the gating risk are each charged exactly once. The clearing discount of Eq. (6) is built on this window.

Publication frequency and dealing frequency. Many issuers publish valuations more often than they deal, a daily publication against monthly dealing being the common case, and the two frequencies are separate inputs. Position health is tested at every published valuation, so a breach can arrive on a publication when no redemption can be submitted. $\mathrm{LH}^{(a)}$ covers that case, so a breach mid-cycle is reserved on the same basis as one arriving on a dealing date. What follows a breach is lending-market behaviour, which the framework takes as given. Controlled operational records state when position health is evaluated and how a breach between dealing dates is treated.

2.2 Worst-case valuation change over the clearing window

The quantity the haircut must bound is the 99% worst-case decline in redemption value of one unit of collateral between the breach and cash settlement, i.e. over the clearing window $\mathrm{LH}^{(a)}$. We do not estimate this from the token's own (short, appraisal-smoothed) valuation history. Instead we inherit from ISDA SIMM v2.6 [1] the prescribed measure of a 99% move over ten business days, restricted to delta margin. The administrator attests a vector of sensitivities s_k in the Common Risk Interchange Format. That file carries DV01 and CS01 ladders, the sensitivities to a one-basis-point move in interest rates and in credit spreads respectively, together with equity and FX deltas, per unit of published valuation. The model uses administrator-attested sensitivities where they are available and applies the conservative proxy rules of Section 3.1.1 otherwise. Each sensitivity is weighted by its published risk weight and concentration risk factor. The five risk classes are interest rate, qualifying credit, non-qualifying credit, equity, and FX, where non-qualifying credit covers securitisations and other non-qualifying names. The weighted sensitivities aggregate within buckets under the published correlations ρ, across buckets under γ with the bucket sums S_b bounded to $[-K_b, K_b]$, and across the five risk classes $r \in \{\mathrm{IR}, \mathrm{CreditQ}, \mathrm{CreditNonQ}, \mathrm{Equity}, \mathrm{FX}\}$ through the inter-class matrix ψ. That roll-up runs inside each SIMM product class p, and the product-class margins then add:

$$\mathrm{WS}_k = \mathrm{RW}_k\, s_k\, \mathrm{CR}, \qquad \mathrm{IM}_p^{(a)} = \sqrt{\sum_{r\in p} K_r^2 + \sum_{r\in p}\sum_{r'\neq r} \psi_{rr'}\, K_r K_{r'}}, \qquad \mathrm{IM}^{(a)} = \sum_p \mathrm{IM}_p^{(a)}, \tag{5}$$

where K_r is the class-level margin built from the WS_k and the product-class margins take no diversification between them (Section 4.5). The full engine is specified in Section 4. Per-type bucket assignment is fixed in Section 3. The scope is delta margin only, and Section 3.1 records why. Sensitivities are attested directly by the administrator, never derived statistically from the token's return history (no principal-component or factor-proxy construction), consistent with the no-fitting principle of Eq. (1). Crypto-native collateral is out of scope entirely (Section 1). An administrator who does not attest a sensitivity does not get zero: unattested rows fall to the Residual bucket at its conservative floor and the parameters freeze against loosening (Sections 2.3 and 4), so refusing to provide Common Risk Interchange Format sensitivities costs LTV, never risk. The model uses the latest valid inputs. Controlled records hold input snapshots and expiry dates. The SIMM parameter set itself is pinned to the published ISDA version.

SIMM is calibrated to a 99% move over ten business days, and $\mathrm{IM}^{(a)}$ enters Eq. (1) verbatim at that horizon, with no rescaling factor applied inside the SIMM engine. The horizon the haircut must cover is the governing clearing window $\mathrm{LH}^{(a)}$, and Eq. (6) takes the 10-day move to it. The inherited standard uses ten business days as its reference horizon. The position is warehoused from the breach through the whole clearing window to cash settlement, $\mathrm{LH}^{(a)}$ business days in total (Eq. (4)). Reconciling the move over that window to the charge at ten business days is an attribution convention, adopted to keep the SIMM core reconcilable against the published standard, and it makes no operational claim about what happens inside the window. The method also defines a stress equivalent under the Basel FRTB liquidity horizons [2]. The controlled reconciliation record holds the per-listing comparison.

The clearing discount δ_{clear}. The clearing discount $\delta^{(a)}_{\mathrm{clear}}$ is the difference between the price risk of the governing window and the charge at ten business days. It is a component of $H^{(a)}$, so the over-collateralisation that funds it sits above the liquidation threshold. The framework sets this reserve and does not prescribe the clearing mechanism: it is silent on how, and by whom, a position in liquidation is cleared, and on how the discount is enforced or paid. Eq. (6) sets the value:

$$\mathrm{IM}^{(a)} + \delta^{(a)}_{\mathrm{clear}} = \begin{cases} \mathrm{IM}^{(a)}\sqrt{\frac{\mathrm{LH}^{(a)}}{10}} & \text{day one (no qualified clearing history)},\\ \max\left(\mathrm{IM}^{(a)}\sqrt{\frac{\mathrm{LH}^{(a)}}{10}},\ \delta^{(a)}_{\mathrm{obs}}\right) & \text{once } \geq N_{\mathrm{obs}} \text{ qualified observations exist}, \end{cases} \tag{6}$$

where the right-hand side is the 99% move over the governing clearing window, and $\mathrm{IM}^{(a)}\sqrt{\mathrm{LH}^{(a)}/10}$ is the SIMM move over ten business days, rescaled to that window by the square root of time. The clearing discount is the residual, $\delta^{(a)}_{\mathrm{clear}} = \mathrm{IM}^{(a)}\big(\sqrt{\mathrm{LH}^{(a)}/10} - 1\big)$ day one. That residual is positive for a window longer than ten business days and negative for a shorter one. A negative value is a margin credit, and it comes from contractually enforceable terms, attested or documented, that fix the window itself (Section 2.1). Estimated legs and FRTB fallback windows never earn one. Money-market collateral dealing daily is the case in point: its attested terms settle in days, and the framework charges the move over those days instead of over SIMM's ten business days. The credit never exceeds the SIMM charge, because $\mathrm{IM}^{(a)} + \delta^{(a)}_{\mathrm{clear}} = \mathrm{IM}^{(a)}\sqrt{\mathrm{LH}^{(a)}/10} \geq 0$ at every admissible window.

The rescaled move is a floor on the total price risk the haircut charges for, never a ceiling: where at least N_{obs} qualified clearing observations exist and their conservative quantile $\delta^{(a)}_{\mathrm{obs}}$

exceeds it, the observed clearing cost governs. $\delta^{(a)}_{\text{obs}}$ is a level and not an increment, the realised clearing cost per unit of published valuation. An observation ageing out of the applicable period can lower $\delta^{(a)}_{\text{obs}}$, and with it the total price-risk charge. A reduction below the rescaled floor requires a shorter attested $\text{LH}^{(a)}$, which lowers the floor itself, or a documented amendment to the framework. The controlled validation record sets the qualification count, quantile, observation period, and diversity requirement.

Adverse evidence tightens the system: observed clearing costs above the prevailing reserve, or repeated failures to clear, force the escalation of Section 5.6. That escalation can widen LH, raise the relevant layer, cut caps, or pause the asset.

The square root of time caveat attaches here. The scaling is exact only for i.i.d. increments, and credit spreads are autocorrelated and jump-prone, so the rescaling is anti-conservative at long horizons for the credit types. The private-credit default add-on layer of Section 2.3 is the explicit compensation there. The securitised and market-neutral fund tails are charged instead through their elevated credit and leverage risk weights, through the rating guardrail's automatic tightening on loss of investment grade, and through the leverage add-on, all of Section 3. The same law rescales the move downward for a window inside ten business days, where a clearing cost that does not shrink with the window would make the credit anti-conservative. The named layers compensate for that risk.

Reasons for the split. The day-one identity of Eq. (6) is the same price-risk calculation, rescaled to the clearing window by the square root of time, so the split costs nothing numerically. It is split out for two reasons. Observed clearing evidence can inform δ_{clear} and nothing else in the haircut, so keeping it distinct from the SIMM core keeps that evidence attributable (the qualified-observation rule of Eq. (6)). IM stays verbatim ISDA SIMM at ten business days, so the validation reconciliation against the published standard holds unchanged.

The same discipline governs missing data. An unavailable input is assigned a prescribed conservative floor in the spirit of FRTB's non-modellable risk factor (NMRF) treatment, and never zero, since zero-filling rewards exactly the assets that produce data voids (Section 4). For an unattested LH, the floor is the documented worst-case window. The practitioner sets a conservative model constant as the floor for each layer that requires one.

The interest cost δ_{carry}. Debt accrues interest for as long as the position is held, so warehousing a position in liquidation through the clearing window has a certain cost that no other component of $H^{(a)}$ charges for. SIMM has no funding-cost risk class, and the clearing discount of Eq. (6) rescales price risk alone, so the interest cost is charged as its own component of the haircut:

$$\delta^{(a)}_{\text{carry}} = \theta^{(a)}\, r^{B,\max}\, \frac{\text{LH}^{(a)}}{252}, \tag{7}$$

where $r^{B,\max}$ is the ceiling of the deployed market's borrow-rate model, the annual percentage yield that model reaches at full utilisation, quoted on a 365-day calendar basis because debt in a decentralised lending market accrues on every calendar day. The deployed market configuration supplies the ceiling as a static input rather than a live utilisation reading. The worked example of Section 4 runs at an illustrative ceiling rate. A kink rate qualifies only where no higher rate is attainable. The window is contracted in business days and spans $\text{LH}^{(a)} \times 365/252$ calendar days, the ratio of the calendar year to the business-day year, so the ceiling is charged over the year fraction $\text{LH}^{(a)}/252$. That share is taken simple rather than compounded, which is the conservative reading of an annual percentage yield at every window inside a year. The rate applies to the debt outstanding when liquidation begins, $\theta^{(a)}$ per unit of published valuation, rather than to the whole collateral value. This is the only place the interest cost is charged: it never enters $\text{IM}^{(a)}$, the liquidation buffer $\nu^{(a)}$, or the gating layer $\lambda^{(a)}_{\text{gate}}$ of Section 2.3.

The threshold $\theta^{(a)}$ itself depends on $H^{(a)}$, so the two resolve together in closed form. Writing $c^{(a)} = r^{B,\max}\,\mathrm{LH}^{(a)}/252$ for the interest cost per unit of debt over the clearing window,

$$H^{(a)} = \frac{\mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \Lambda^{(a)} + \kappa^{(a)} c^{(a)}}{1 + \kappa^{(a)} c^{(a)}}, \qquad \theta^{(a)} = \kappa^{(a)}\big(1 - H^{(a)}\big), \qquad \delta^{(a)}_{\text{carry}} = \theta^{(a)}\, c^{(a)}. \tag{8}$$

The numerator holds every non-interest component of $H^{(a)}$: a type with a deterministic fee addend books it inside $\delta^{(a)}_{\text{clear}}$, so the identity $\delta^{(a)}_{\text{carry}} = H^{(a)} - H^{(a)}_0$, with $H^{(a)}_0$ the haircut before the interest cost, is exact.

Each period of interest accrual is reserved exactly once. The liquidation buffer $\nu^{(a)}$ charges for the borrow rate before liquidation begins, across one interval between published valuations, and it protects the borrower from being pushed through $\theta^{(a)}$ by routine drift (Eq. (11)). The interest cost $\delta^{(a)}_{\text{carry}}$ charges for the same rate after liquidation begins, across the clearing window, and it protects the deployment's lenders while the position is warehoused. The two intervals do not overlap and the two parties differ. Neither term enters $\lambda^{(a)}_{\text{gate}}$, which charges for the chance that a clearing window does not pay and the cost of the extension that follows, never the certain interest cost over the contractual window.

2.3 Additive layers for structural blind spots

The sensitivity-based core models smooth co-movement of modelled risk factors. Everything it structurally cannot see enters as a named additive layer. The total layer charge is

$$\Lambda^{(a)} = \sum_j \lambda^{(a)}_j, \tag{9}$$

a sum of fixed percentages per type. The practitioner sets the model constants used to determine these values (Appendix A). Each λ_j is tied to a specific structural blind spot:

(i) **Gating/suspension (λ_{gate}).** SIMM has no liquidity risk class: the narrowing of the exit (gates, side-pockets, suspensions) has no risk factor. This is the central tail risk of lending against redemption-routed collateral. Funds gate when redemption is most needed.

(ii) **Valuation staleness/smoothing.** Sensitivities are computed off administrator-published valuations that are smoothed relative to traded prices. Risk weights calibrated on traded histories partially compensate (a strength of inheritance), but the residual exposure between publications needs its own charge.

(iii) **Default add-on (private credit collateral).** SIMM margins spread delta rather than jump-to-default. FRTB pairs its sensitivity charge with a default risk charge for precisely this reason. A $PD \times LGD$ add-on in the FRTB-DRC style covers the gap.

(iv) **Oracle pipeline integrity.** The valuation oracle is a data pipeline with no SIMM risk factor. A weak configuration, such as a manual or single-signed feed, has its own charge by configuration tier, zero at the Robust tier.

(v) **Prepayment (amortising consumer collateral).** Borrower prepayment optionality on amortising consumer collateral shifts cashflow timing in a way the delta margin alone does not charge for. SIMM's native compensator would be vega and curvature margin, but those require attested implied-volatility sensitivities, and no options trade on non-traded consumer collateral from which to attest them. The framework instead evaluates tranche CS01 at the slower of the disclosed conditional prepayment rate and a stressed-extension floor, and books the residual optionality as this named layer (the ¶ 9 substitution of Section 3.1; Section 3.1.1).

(vi) **Manager/operational.** Wrapper, transfer-agent, and administrator failure modes have no market sensitivity.

(vii) **Wrong-way.** Co-realisation of collateral stress with the stress of the borrower base (e.g. private-credit borrowers who are crypto market-makers) is not a SIMM object.

(viii) **Leverage add-on (leveraged fund collateral).** Fund-level leverage scales tail risk beyond what the fund-unit risk weight reserves. The add-on is charged per turn of leverage above policy.

(ix) **Event risk (event-linked collateral).** A collateral whose dominant loss is a jump with no SIMM risk factor at all, such as an insurance catastrophe event, takes a named event layer set from attestable model documents. Event probability grows roughly linearly in exposure time with undiminished severity, so neither the margin over ten business days nor the square-root-of-time extension can hold it, and the layer is the designated home.

The decomposition $H = \mathrm{IM} + \delta_{\text{clear}} + \delta_{\text{carry}} + \Lambda$ keeps every charge attributable: a reviewer can challenge any single number without reopening the rest. The final haircut is thus D-SIMM. The risk-factor reference of the supporting documentation records, factor by factor, the metric behind each of these charges, its evidence source, and the parameter it drives.

2.4 Policy limits

The liquidation threshold and borrow limit follow from (1):

$$\theta^{(a)} = \kappa^{(a)}\bigl(1 - H^{(a)}\bigr), \qquad \ell^{(a)}_{\max} = \theta^{(a)} - \nu^{(a)}, \qquad \nu^{(a)} \in [0.04,\, 0.08]. \tag{10}$$

The per-listing liquidation buffer $\nu^{(a)}$ is the *borrower-side* gap between the borrow limit $\ell^{(a)}_{\max}$ and the liquidation threshold $\theta^{(a)}$. The 4–8pp range in (10) is a fixed policy constant. Per-listing values within it are set at listing (Section 5.1).

Solvency by construction. Liquidation begins when the loan-to-value ratio crosses $\theta^{(a)}$, so the over-collateralisation cushion protecting the system at the point of liquidation is $1 - \theta^{(a)} \geq H^{(a)} = \mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \delta^{(a)}_{\text{carry}} + \Lambda^{(a)}$, with equality at the Prime tier ($\kappa = 1$). Whatever discount compensates the party that holds the position to the redemption point is paid out of this cushion. The discount is bounded by $\mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \delta^{(a)}_{\text{carry}}$, and the cushion $1 - \theta^{(a)} \geq H^{(a)}$ is available above the liquidation threshold. The layer total $\Lambda^{(a)}$ remains once the bound is exhausted, strictly so wherever the layer total is positive. Day one, before any qualified clearing observation governs, a window inside ten business days makes the price-risk charge $\mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}}$ smaller than the verbatim figure at ten business days. That is the intended reading of an asset reaching cash inside SIMM's ten-business-day horizon, and the interest cost $\delta^{(a)}_{\text{carry}}$ is charged separately.

Placed below the liquidation threshold $\mathrm{LTV} = \theta^{(a)}$ instead, in the gap between the borrow limit and the liquidation threshold, the clearing discount could draw only on the buffer $\nu^{(a)}$ at the point of liquidation. That buffer is smaller than $\delta^{(a)}_{\text{clear}}$ wherever the margin charge is large and the clearing window long, as for a bucket-11 fund unit at the published SIMM risk weight for that bucket (19 at v2.6) over a 45 to 95 business-day window. That design would be insolvent at the point of liquidation for exactly the assets that need the discount most.

Between two usable published valuations a position's debt accrues at the borrow rate while the collateral oracle accrues at the attested yield $y^{(a)}$, and at the next publication the collateral may jump by $J^{(a)}$. A borrower at exactly $\ell_{\max}$ must not be pushed through θ by that interest accrual plus one ordinary adverse valuation publication, which requires

$$\nu^{(a)} \;\geq\; \ell^{(a)}_{\max}\, \frac{r^{B,\max}\, \Delta t^{(a)}_{A}/252 + \bigl|q_p\bigl(J^{(a)}\bigr)\bigr|}{1 - \bigl|q_p\bigl(J^{(a)}\bigr)\bigr|}, \tag{11}$$

where $r^{B,\max}$ is the ceiling borrow rate of the deployed interest-rate model. The inequality is exact. Debt grows by the factor $1 + r^{B,\max}\Delta t^{(a)}_{A}/252$ over the interval while the published valuation the ratio is measured against falls by $\bigl|q_p(J^{(a)})\bigr|$, so the requirement takes the denominator

$1-|q_p(J^{(a)})|$. Dropping that denominator understates the floor by exactly $|q_p(J^{(a)})|$ times the floor itself. At a 15% floor it is 2.7 points, against a buffer of 4 to 8 points. The interval $\Delta t_A^{(a)}$ is the longest gap between usable published valuations in business days, a usable valuation being one the deployment's valuation oracle can act on. It is the attested publication frequency, the publication cycle $\Delta^{(a)}$ of Section 2.1 counted in business days, plus the attested publication lag, and the debt accrues across it on the 365-day calendar basis of Eq. (7). The requirement takes no credit for the attested yield $y^{(a)}$: the published-valuation rebase and the accrual of interest on the debt run on different frequencies, so a yield offset can be too large or too small at any given time, and neither error is safe. The requirement is checked per listing in the controlled buffer record (Section 5.1), and the adopted constants must dominate it there.

The jump term $q_p(J^{(a)})$, the lower p-quantile of the jump between consecutive published valuations, is a *prescribed* per-type stressed-jump floor. It is policy, never fitted: the framework bans estimating it from the token's own valuation history. The practitioner sets the stressed-jump floor and quantile level p as model constants (Appendix A). Eq. (11) motivates the size of $\nu^{(a)}$ and is never evaluated on live data. The liquidation buffer is the static per-listing constant of Section 5.1. Types with infrequent valuation publication and large jumps between published valuations take the wide end of the range, and types with daily published valuations the narrow end. A type whose prescribed floor exceeds its adopted buffer launches under the new-borrow guardrail of Section 5.5 rather than with a wider buffer. The liquidation buffer $\nu^{(a)}$ absorbs routine dynamics. The haircut $H^{(a)}$ is the reserve against stress. The two are not interchangeable, which is why (10) subtracts them in sequence rather than blending them.

Eligibility and tiering sit upstream of (10): the practitioner's due-diligence scorecard screens listing through eligibility failures and assigns a tier multiplier $\kappa \le 1$ from the practitioner's own tier structure (illustratively $\{1.00, 0.90, 0.75\}$ for Prime/Standard/Restricted). A bridge or cross-chain dependence is not one of those failures. It places the asset outside the framework's scope, and the asset is not scored at all (Section 1.1). Lower tiers hold the larger over-collateralisation cushion $H_{\text{eff}} = 1-\theta \ge H$. The borrow limit follows as $\ell_{\max} = \theta - \nu$, with the liquidation buffer ν applied as an absolute deduction rather than scaled by κ. Section 5 states the public policy constraints. A fully worked composition of (1) appears in Section 4.

2.5 Capacity and caps

The capacity reference, the single attested quantity from which both caps are set, is $R_{\text{cap}}^{(a)}$, the issuer's attested redemption capacity in USD per dealing window. It is applied below and in Sections 5–6 as the one-window capacity, the volume the issuer redeems within one dealing window. Where the dealing window differs from the basis on which the capacity is attested, the attested figure is window-normalised before it is used anywhere, because a capacity attested over a period longer than the dealing window overstates what one window can clear. The window basis is part of the definition of $R_{\text{cap}}^{(a)}$ rather than a free parameter (Appendix A). Both caps are closed-form functions of it:

$$B_{\max}^{(a)} \;\le\; R_{\text{cap}}^{(a)}, \qquad S_{\max}^{(a)} \;\le\; M\, R_{\text{cap}}^{(a)}, \tag{12}$$

where M is the supply-cap multiple: accepted supply is capped at M dealing windows of attested capacity, a free parameter of the practitioner (Appendix A). $B_{\max}^{(a)}$ caps borrowed *debt* against collateral a at the pool level and is denominated in units of borrowed debt, while $R_{\text{cap}}^{(a)}$ and $S_{\max}^{(a)}$ are denominated in USD of collateral valued at the published valuation. The liquidator redeems collateral at the published valuation only until the debt is repaid, so the volume one dealing window must clear for the largest defaulting borrower is the *debt* $B_{\max}^{(a)}$. The posted collateral above the repaid debt is redeemed later at the borrower's risk. Enforced at the pool level, the cap $B_{\max}^{(a)} \le R_{\text{cap}}^{(a)}$ therefore guarantees that one window clears the pool's entire debt at the point of liquidation, and the interest accruing over the window is reserved separately by $\delta^{(a)}_{\mathrm{carry}}$. The supply cap bounds the aggregate portfolio in the same collateral units so that accepted supply clears within M windows of attested capacity.

Control of the caps. The deployment exposes caps per collateral rather than per entity, and the framework assumes no ability to identify a borrower: addresses are cheap, and nothing prevents one economic actor from borrowing through many of them. The enforced controls are therefore the per-collateral parameters a deployment may natively support: the supply cap $S^{(a)}_{\max}$, the borrow cap read at the pool level, and the ceiling $\ell^{(a)}_{\max}$. The deployed market configuration shows both active caps. Controlled records hold each recomputation from $R^{(a)}_{\mathrm{cap}}$ and the selected constant M. These support a guarantee that holds however debt is split across borrowers. Where the venue enforces the pool-level borrow cap at $B^{(a)}_{\max} \leq R^{(a)}_{\mathrm{cap}}$, one dealing window clears the pool's entire debt at the point of liquidation. Where it enforces only the supply cap, total debt is at most $\ell^{(a)}_{\max} S^{(a)}_{\max} < M\, R^{(a)}_{\mathrm{cap}}$, so a full redemption completes within M windows of dedicated capacity. The per-entity reading of $B^{(a)}_{\max}$, that the largest single borrower's debt clears in one window, is a monitored diagnostic rather than an assumption. Where the deployment cannot attribute debt to entities, the framework claims only the pool-level guarantee, which requires no identification of anyone.

Remark. The execution price does not vary with the quantity redeemed. Under redemption at the published valuation, the execution price of a redemption of size q is the published valuation at settlement, constant in q up to R_{cap}, so $\partial P_{\mathrm{exec}}/\partial q = 0$. The mechanism that couples $\ell_{\max}$ to $S_{\max}$ in impact-based frameworks therefore has identically zero gain here: in those frameworks a larger portfolio forces a larger stress exit, which moves price further, which lowers the viable LTV, and that price move is absent under redemption at the published valuation. The remark is independent of the caps: $\partial P_{\mathrm{exec}}/\partial q = 0$ is a property of the redemption process alone, so the one-directional computation remains valid with per-entity concentration uncontrolled, as it is here. Caps and CR bound the time to exit and haircut steepness, never the exit price, so weakening them lengthens the feasibility horizon of Section 6.2 ($\lceil B_{(1)}/R_{\mathrm{cap}} \rceil$ windows to clear the largest debt instead of one) without reintroducing a circular dependency. The bound $\mathrm{CR} \leq \sqrt{M}$ rests on the aggregate supply cap $S_{\max}$, a native parameter of the deployment unaffected by how per-entity concentration is controlled.

At each recomputation, the attested inputs of Eq. (1) give $\mathrm{IM}^{(a)}$, $\delta^{(a)}_{\mathrm{clear}}$, $\Lambda^{(a)}$, and $\delta^{(a)}_{\mathrm{carry}}$, then $H^{(a)}$, $\theta^{(a)}$, and $\ell^{(a)}_{\max}$. The attested capacity $R^{(a)}_{\mathrm{cap}}$, with the policy constant M, gives the borrow cap and the supply cap of Eq. (12). No output is an input to its own calculation. CR is evaluated at the observed exposure, an input read from the current state rather than solved jointly with the limits. The re-keyed threshold $T = R^{(a)}_{\mathrm{cap}}$ and the bound $\mathrm{CR} \leq \sqrt{M}$ are derived in Section 4.

The borrow rate does not reopen a loop. The ceiling rate $r^{B,\max}$ is a static constant of the deployment, and the interest cost it reserves resolves in closed form at Eq. (8). The rate a borrower actually pays appears only in the motivation for the buffer (11), so a utilisation-dependent borrow rate feeds no live quantity dependence into the computation.

3 Risk factors and how each is charged

The framework fixes the risk factors and their treatments, and the practitioner defines their own collateral types against them. Type criteria are qualitative, and the practitioner sets them. Each criterion rests on attested facts read from offering documents and fact sheets. Where no type fits, the practitioner defines a new type against the fixed risk factors. Existing criteria remain unchanged.

Every retained factor feeds exactly one component of Eq. (1). The four haircut components are the market-risk core $\mathrm{IM}^{(a)}$, the clearing discount $\delta^{(a)}_{\mathrm{clear}}$ (Section 3.3), the interest cost $\delta^{(a)}_{\mathrm{carry}}$, and the named additive layers $\Lambda^{(a)}$. The interest cost takes no risk factor because it is a certain cost, the interest the debt accrues at the ceiling borrow rate for as long as the position is held (Eq. (7)). The route from each component to the on-chain limits is set out in Sections 4 and 5.

The scope conditions of Section 1 govern throughout.

3.1 Market

The market-risk core is the delta component of the ISDA SIMM v2.6 initial margin. It uses the standard's published risk weights, correlations, and ten-business-day horizon unchanged [1]. The prescribed numbers are inherited as published, and the deviations recorded here are deviations of scope and granularity rather than of any prescribed value. Administrator-attested Common Risk Interchange Format sensitivities s_k enter the weighted sensitivities of Eq. (19), each scaled by its published risk weight RW_k and by the concentration risk factor $\mathrm{CR}^{(a)}$ of Section 3.3. The weighted sensitivities are then combined within each bucket under published correlations, across buckets, and finally across the risk classes, under the published SIMM correlations at each step. The ψ roll-up across the risk classes IR, CreditQ, CreditNonQ, Equity, and FX uses the published inter-class matrix unchanged, applied inside each SIMM product class, and the product-class margins then add with no diversification between them (Section 4.5).

SIMM ¶ 9 attaches vega and curvature margin only to instruments that are options or embed one, including prepayment options. A type that is a linear, long-only holding of fund units (the fund shares the tokens represent, Section 3.1.2) attracts neither. Amortising securitised exposures are the exception. Consumer-credit tranches and mortgage-backed sleeves embed prepayment optionality, and they would attract vega and curvature under a strict ¶ 9 reading. The framework instead substitutes the tranche CS01 evaluated at the slower of the disclosed conditional prepayment rate and a stressed-extension floor (Section 3.1.1) plus the named prepayment layer. The practitioner sets its per-type magnitude as a model constant. That substitution is documented and conservative rather than a SIMM waiver. It is recorded as assumption A14 of the assumptions addendum published alongside this paper.

Cross-currency basis and sub-curve granularity are dropped. A sub-curve is retained wherever a floating-rate type's coupons reset off it (SIMM ¶ 14), so that type's DV01 is recorded on that curve. Base correlation applies only to CDX/iTraxx index-tranche positions, which are out of scope, so it never enters.

$\mathrm{IM}^{(a)}$ is the 99% move over SIMM's native ten-business-day horizon, computed from *prescribed* shocks. A position in liquidation is held for the whole clearing window $\mathrm{LH}^{(a)}$ instead (Section 2.2). $\mathrm{IM}^{(a)}$ is never estimated from the token's own appraisal-smoothed valuation history, whose naive volatility estimate is biased low. The clearing discount $\delta^{(a)}_{\mathrm{clear}}$ of Section 2.2 rescales that move to the clearing window, in either direction.

3.1.1 SIMM bucket assignment

Each administrator-attested Common Risk Interchange Format row names its own risk class, bucket, and qualifier under SIMM v2.6's own definitions. Those definitions cover IR vertices (¶ 14), CreditQ issuer/seniority risk factors (¶ 15) bucketed by quality and sector (¶ 38), CreditNonQ tranche risk factors (¶ 16) with their buckets (¶ 45), equity fund units (¶ 17), and the qualifying-securitisation test (¶ 24). Each listing's assignment is fixed at onboarding as a documentation exercise, re-verified each valuation cycle, and any unattested row drops automatically to the Residual penalty route (Section 4.6).

The assignment is derived, and the day-one sensitivity values behind it are estimates rather than attested numbers. They are built from the issuer's offering documents under the conservative proxy rules stated here. Those rules take the whole duration at the nearest longer vertex, one name in the bucket with the highest risk weight, and the slower of the disclosed and the stressed prepayment speed. Each proxy is constructed to bound the attested figure from above, and the administrator-attested Common Risk Interchange Format sensitivities file replaces the estimates at onboarding and at each valuation cycle after it. A fund-unit type whose driving input is the fund-unit delta itself is the exception, because that delta is 100% of the published valuation, a documented fact rather than an estimate. An event-risk layer magnitude, by contrast, is a day-one estimate pending the attested event-model file.

3.1.2 Fund-unit treatment for market-neutral funds

A market-neutral portfolio's long and short legs correspond to the same SIMM risk factors, so net look-through Common Risk Interchange Format sensitivities are ≈ 0: naive look-through certifies a leveraged basis portfolio as riskless precisely because its real risk (a widening of the basis, a funding squeeze, forced deleveraging) lives in the residuals SIMM nets away.

Per SIMM ¶17, market-neutral fund collateral is therefore margined as Equity bucket-11 fund units at the published SIMM risk weight for that bucket (19 at v2.6). The look-through Common Risk Interchange Format file is used for *validation only, never netting.*

The treatment requires a valid administrator-signed neutrality and leverage file. Non-attestation reclassifies the position to Equity Residual (Section 4.6). The controlled data record sets the refresh requirement.

3.1.3 FX share-class hedging

SIMM FX risk applies only where the share class is not fully hedged to the borrow currency, and the condition is generic: for *any* type whose share-class or base currency differs from the borrow currency, an administrator attestation of the hedge ratio is a *hard eligibility condition* rather than a haircut input. For example, for a daily-dealing bond fund, an unhedged EUR share class adds the SIMM v2.6 FX risk weight on the full published valuation, by itself enough to push a type outside its model limits. For a fund reached through a fund-of-funds wrapper, the attestation covers the whole wrapper stack, since the share-class label does not resolve the realised exposure. Where the look-through of Section 3.2.4 finds holdings stated outside the borrow currency, their rate and spread sensitivities are recorded on that currency's own curve and the residual currency exposure enters the FX risk class, never merged into the USD vertices, since SIMM's interest-rate buckets are per currency.

3.1.4 Valuation staleness

For a tokenised asset with no traded market the published valuation is the only valuation there is, so no deviation between an oracle and a market price exists to charge for. The exposure is age. The asset's value moves between publications, and nobody can trade against an out-of-date publication on a transfer-restricted token, so the whole error sits with the deployment. A borrower can draw at full LTV against an optimistic published valuation that is one publication out of date.

The layer has two parts. The baseline $\lambda_{\text{stale}}^{\text{base}}$ is a model constant set by the practitioner for each type and covers the ordinary age of an on-time publication. It remains in force while publication stays within the contractual publication cycle. The public worked example uses the illustrative value $\lambda_{\text{stale}}^{\text{base}} = 0.5\%$. A type valued by appraisal additionally has a fixed smoothing component inside that baseline. An escalation $\Delta\lambda_{\text{stale}}^{(a)}$ is added while publication is overdue, so the layer in force is $\lambda_{\text{stale}}^{(a)} = \lambda_{\text{stale}}^{\text{base}} + \Delta\lambda_{\text{stale}}^{(a)}$.

The escalation is proportional to the volatility implied by the type's published SIMM risk weight:

$$\Delta\lambda_{\text{stale}}^{(a)} = \hat{\sigma}^{(a)}\sqrt{\Delta_{\text{stale}}^{(a)}}, \qquad \hat{\sigma}^{(a)} = \mathrm{RW}^{(a)}\,\frac{\sqrt{365/14}}{\Phi^{-1}(0.99)}, \tag{13}$$

where $\Delta^{(a)}_{\text{stale}}$ (in years) is the elapsed time since the last official publication in excess of the contractual publication cycle plus publication lag, floored at zero, and $\text{RW}^{(a)}$ is the type's dominant risk weight. No fitted volatility enters. The construction uses the published SIMM risk weight in two steps. $\text{RW}^{(a)}$ is the prescribed 99% move over ten business days, so $\text{RW}^{(a)}/\Phi^{-1}(0.99)$ is the standard deviation of that move on a Gaussian reading. The factor $\sqrt{365/14}$ then rescales that standard deviation from ten business days, read as 14 calendar days, to one year. $\hat{\sigma}^{(a)}$ is therefore an annualised volatility, and $\hat{\sigma}^{(a)}\sqrt{\Delta^{(a)}_{\text{stale}}}$ is the one-standard-deviation move over the overdue interval. The Gaussian back-solve and the reading of business days as calendar days are recorded as assumption A11.

Charging one standard deviation rather than a second 99% quantile is deliberate. The stressed move across one ordinary interval between publications is absorbed by the liquidation buffer $\nu^{(a)}$, which covers one adverse publication at the prescribed jump floor (Eq. (11)). Disclosure quality is charged through the due-diligence multiplier κ instead (Section 5.2).

Oracle pipeline integrity (λ_{oracle}). Staleness charges for an *old but trusted* valuation. It does not charge for the risk that a *fresh* one is wrong: a mispublished, single-sourced, unsigned, or manipulated published valuation, the recurring traditional-finance data-pipeline failure. That risk is not spread evenly across the collateral, and the layer $\lambda^{(a)}_{\text{oracle}}$ charges for the difference. $\lambda^{(a)}_{\text{oracle}}$ is set by the asset's *effective* oracle tier, the lower of its configuration tier and the provider-fitness cap. That tier reaches Robust, $\lambda_{\text{oracle}} = 0$, only where the feed has source multiplicity, a signed payload, an enforced max-age circuit breaker, and an unaffiliated operator. Single-signed and manual tiers, and issuer-affiliated providers, take a prescribed positive charge.

Where a position is held against a claim on a redemption awaiting settlement, no external feed publishes a value for that claim: it is valued by the deployment's own computation from the last published valuation, so it has neither the source multiplicity nor the signing that earns the Robust tier. That is the case to charge conservatively, and the practitioner sets the effective tier and the charge for a valuation with no independent published feed behind it (free parameter, Appendix A).

Dual attestation and a configured plausibility check are *tier predicates that earn the Robust tier* ($\lambda_{\text{oracle}} = 0$) rather than separately-charged controls. A breach found by that check is monitor-only. It never produces an automatic charge or tier change. Oracle configuration is charged once, in λ_{oracle}. Its effect on the transparency score is an eligibility requirement rather than a second charge, and no scorecard score acts on λ_{stale} (Section 5.2).

3.2 Credit

Spread risk already sits in the market-risk core as CreditQ/CreditNonQ delta. The credit factor charges for what spread delta cannot see: jump-to-default and rating migration (legal structure is screened at eligibility rather than charged).

3.2.1 Default add-on (private credit only)

FRTB pairs the sensitivity charge with a default risk charge for exactly this reason, and the framework mirrors that pairing as an additive layer:

$$\lambda^{(a)}_{\text{def}} = \text{PD}\big(r^{(a)}, \text{LH}^{(a)}\big) \times \text{LGD}^{(a)}, \tag{14}$$

with PD taken from rating-agency cohort default tables at rating $r^{(a)}$, horizon-scaled to the clearing window $\text{LH}^{(a)}$ (4), and LGD from published senior-secured recovery tables rather than from CDS, which do not trade on these issuers. Senior private credit admits unrated borrowers, so $r^{(a)}$ is resolved by a prescribed proxy hierarchy rather than omitted: an issuer or facility rating from a nationally recognized statistical rating organization where one exists, else a manager internal rating corroborated by an independent party (auditor or administrator), else the prescribed unrated-sector PD floor. Eq. (15) states that hierarchy as the definition of $r^{(a)}$:

$$r^{(a)} = \begin{cases} r_{\text{agency}} & \text{a nationally recognized statistical rating organization rates the issuer or facility,} \\ \min\bigl(r_{\text{int}} - 1 \text{ grade},\ \text{BB+}\bigr) & \text{else a corroborated internal rating exists,} \\ r_{\text{floor}} & \text{else,} \end{cases} \tag{15}$$

where grades are ordered by credit quality and min returns the weaker of the two. A corroborated internal rating therefore resolves at BB+ or weaker, and no internal rating resolves to investment grade.

The internal rating r_{int} is produced by the controlled look-through scorecard over the credit dimensions of the due-diligence questionnaire: portfolio-weighted leverage, interest cover, seniority, collateralisation, and delinquency. The scorecard output is expressed on the agency scale. A corroborated internal rating is reduced by one grade because the manager of the collateral produced it. The practitioner sets the floor grade r_{floor} as a model constant. The window dependence then follows from Eq. (16) and the seniority dependence sits in LGD rather than in PD. The floor is weaker than the BB+ cap, so weaker evidence never produces a better parameter than stronger evidence.

The horizon scaling of Eq. (14) is the survival form of a constant default intensity:

$$\text{PD}\bigl(r, \text{LH}\bigr) = 1 - \bigl(1 - \text{PD}_1(r)\bigr)^{\text{LH}/252}, \tag{16}$$

where $\text{PD}_1(r)$ is the published one-year cohort default rate at grade r and 252 is the business-day year. At every window shorter than a year the scaled PD sits slightly above the linear pro-rata value $\text{PD}_1(r) \cdot \text{LH}/252$, because $1 - (1-p)^t$ is concave in t and agrees with the linear value at $t = 0$ and $t = 1$. That is the conservative direction. Where the governing clearing window reaches a year, PD is read from a multi-year cohort column instead.

Default clustering is the one direction in which the form understates default risk. Defaults arrive in stressed cohorts rather than at a constant rate, so a window inside a stressed year contains more default risk than a constant intensity implies. The one-grade reduction on corroborated internal ratings and the floor grade are the prescribed offsets against that. The scaling assumption and that caveat are recorded as assumption A13 of the assumptions addendum.

$\text{LGD}^{(a)}$ is read from a published senior-secured recovery table keyed to attested seniority. Where seniority is unattested the table cannot be keyed, and LGD is floored at the prescribed LGD values of the FRTB default risk charge: 75% for senior debt and 100% for non-senior debt or an equity instrument (MAR22.12) [2]. In that case the claim takes the non-senior value, 100%, because a floor takes the worst admissible reading rather than the more favourable one. The prescribed value replaces the recovery-table value rather than averaging with it. The unattested-seniority floor is the prescribed non-senior value of 100%.

Where no admissible rating and no seniority attestation can be obtained, Eq. (14) cannot be parameterised from evidence, and the framework prescribes the outcome. The default add-on takes the floor grade and the LGD floor, $\lambda_{\text{def}}^{(a)} = \text{PD}\bigl(r_{\text{floor}}, \text{LH}^{(a)}\bigr) \times \text{LGD}_{\text{floor}}$, and the asset is capped at the practitioner's lowest eligible tier (Section 5.2). Both consequences apply automatically once the evidence is absent, in the same way as every other missing input (Section 4.6).

The model constant set by the practitioner for each type is a floor on the derived value:

$$\lambda_{\text{def}}^{(a)} = \max\Bigl(\text{PD}\bigl(r^{(a)}, \text{LH}^{(a)}\bigr) \times \text{LGD}^{(a)},\ \lambda_{\text{def}}^{\text{floor}}\Bigr), \tag{17}$$

where $\lambda_{\text{def}}^{\text{floor}}$ is the model constant set by the practitioner for the type. The layer takes a positive value within the private-credit layer stack and is identically zero elsewhere. The collateral types assign every other default tail elsewhere. A short-duration or investment-grade tail is negligible at these horizons, a securitised tail is charged through its non-qualifying risk weight plus the rating guardrail below, and a leveraged-fund tail through its fund-unit risk weight plus the leverage add-on (Section 3.4). An event-linked tail is charged through the named event-risk layer, a deliberate choice over a PD×LGD add-on, since the instrument's default *is* the insured event.

3.2.2 Event risk

Some collateral has a dominant loss that arrives as a jump with no SIMM risk factor behind it, an insurance catastrophe event being the standard case. The probability of such a jump grows roughly linearly in exposure time, and its severity does not diminish as the horizon shortens. Neither the margin over ten business days nor the square-root-of-time extension of Eq. (6) holds that shape, so the charge sits in the named event layer $\lambda^{(a)}_{\text{event}}$ instead.

The layer is set from attestable model documents rather than fitted. It takes the larger of the attested 99% annual-aggregate modelled valuation loss and the largest attested modelled single-event valuation impact, both read from the manager's event-model file. Where the file is unattested or uncorroborated the layer takes a prescribed conservative floor. Where the collateral type assigns its dominant loss to this layer, the default add-on of Eq. (17) is zero for that type.

3.2.3 Structure requirement

Bankruptcy remoteness, segregated custody, independent administrator and auditor, and enforceable redemption rights are scored as the legal-structure and governance dimensions of the due-diligence scorecard. The structure score *screens eligibility* rather than feeding the haircut. The resulting tier applies a multiplier to the limits for new exposure (illustratively Prime 1.00, Standard 0.90, Restricted 0.75). The tier acts on the over-collateralisation cushion only (Section 5.2). Eligibility failures (legal-core gaps, unresolved redemption gates, material restatements, live sanctions or material enforcement, single-key control, bad-faith refusal, and unrecovered exploits) precede scoring. An asset with a bridge or cross-chain dependence sits outside the framework's scope and is not scored at all. The conditional failures reject only the *unmitigated* state, while live sanctions reject in every state.

3.2.4 Look-through from fund documents, and rating guardrails

The sensitivities SIMM consumes are derived by look-through from the issuer's own documents wherever no attested sensitivities file exists. Where the asset is a feeder token, the same reading runs through to the master fund: the practitioner reads the master's offering memorandum and investment limits, its current factsheet, and its latest administrator or trustee report, and the master's administrator is the attesting party for the master-level legs. For a credit-routed type that reading also fixes the CreditQ bucket assignment. From the issuer's own documents the practitioner states approximate holdings: portfolio size, the stated sector and seniority mix, the stated term or weighted-average life, whether coupons float, and the reset frequency. Each holding group is assigned its SIMM risk factors from those facts, an interest-rate vertex from its stated reset or life, a CreditQ bucket from its stated quality and sector, and a CS01 from spread duration times composition weight. The weighted sensitivities of Eq. (19) and the published aggregation then give the approximate exposure the margin charges.

The same procedure produces the sensitivity inputs of every type whose margin routes through the interest-rate and credit risk classes. The fund-unit types take their margin input from the fund-unit delta instead (Section 3.1.2).

Inferred sensitivities and the recorded reason. Where the administrator delivers a DV01 or a CS01, the delivered figure is used. Where it does not, the sensitivity is inferred, and the inference is recorded with the document line it rests on and the reason for the reading taken. An attested 90-day revolving portfolio is the standing example. A 90-day revolving term is short-dated credit exposure, so the spread sensitivity is recorded at the shortest SIMM credit vertex and the rate sensitivity at the reset vertex. Documented extension or amendment practice that keeps facilities outstanding for years displaces that reading and moves the sensitivity to a longer vertex.

Every inferred input takes a conservative proxy that bounds the attested figure from above under the SIMM bucket-assignment rules of Section 3.1.1. An inference can therefore lower the permitted LTV and never lowers the haircut. The practitioner sets the standard each inference is held to, which readings are admissible, and what displaces a reading already recorded (free parameter, Appendix A).

Replacement of inferred inputs. Delivered evidence replaces conservative proxies input by input. A delivered sector tag takes its own bucket, while any untagged remainder keeps its recorded proxy. The framework restores the recorded proxy when a tag becomes stale. An exposure the documents leave unclassifiable falls to Residual at full marginal cost (Section 4.6).

Rating guardrails (securitised collateral). The actions a loss of investment grade forces, the evidence that confirms the loss, and the observation window it is measured over are stated in Section 5.5. For a fund type holding rated securitised sleeves the same guardrail applies per sleeve: a lapsed sleeve rating attestation drops that sleeve to Residual, and *confirmed* loss of investment grade on a sleeve reassigns it to CreditNonQ bucket 2 with the identical mechanics at escalation levels 3 and 4.

3.3 Redemption liquidity

The exit is contractual: redemption from the issuer, not a sale into a market. Liquidity risk on this universe is the risk that the redemption process *lengthens* (notice periods, queues, partial fills) or *narrows* (gates, suspensions), and that positions outgrow attested capacity. Every input is read from offering documents or attested transfer-agent data at existing reporting frequency. Nothing is fitted.

3.3.1 The clearing window and the clearing discount

When liquidation begins at LTV $= \theta^{(a)}$, the position is held through the clearing window to cash. The window is $\mathrm{LH}^{(a)}$ of Eq. (4), the worst-case wait to the next submission deadline plus the attested notice, dealing, and settlement legs, fixed by the three-source hierarchy of Section 2.1.

The window sets the day-one value of the clearing discount $\delta^{(a)}_{\text{clear}}$ of Eq. (6), an *additive haircut component sitting above the liquidation threshold* rather than a gap below it. The practitioner calculates each listing's day-one values from its attested terms (Section 2.1).

Listing requires no committed settlement level beyond the subscription documents' own terms. A documented or attested committed settlement level bounds the borrow cap through the committed-settlement term of Eq. (35). It also shortens $\mathrm{LH}^{(a)}$ (Eq. (4)), a reduction earned by evidence against the floor state in which no party stands ready at the moment of need (Section 6.3).

3.3.2 Gating and suspension provisions

Gate thresholds (% of the published valuation per dealing date), suspension discretion, side-pocket rights, and holdbacks are extracted from offering memoranda and assigned to the gating and suspension layer λ_{gate}. The dominant tail of lending against a published valuation is that funds gate exactly when the deployment needs to redeem, and no SIMM risk factor can charge for it. A *live, unresolved, or disorderly* gate, suspension, or side-pocket is an eligibility failure. A *resolved, as-designed* pro-rata gate that made investors whole is not (it caps the redemption-firmness dimension of the practitioner's scorecard).

3.3.3 Queue and pro-rata mechanics

Where the offering documents provide pro-rata scale-back, gated redemptions fill pro-rata rather than first-come-first-served, and partial fills roll the unfilled balance into subsequent dealing windows and lengthen the *realised* clearing window beyond the contractual one. Transfer agents report queue depth and gate utilisation. Sustained utilisation lengthens the realised $\mathrm{LH}^{(a)}$, recomputes the clearing discount (6) on that window, and escalates per the gating guardrail.

3.3.4 Attested redemption capacity, concentration, and caps

The single governing capacity number is $R^{(a)}_{\text{cap}}$, the redemption capacity per dealing window, attested by the issuer or transfer agent. It re-keys the SIMM concentration threshold, the dealer-market T_b replaced by $T^{(a)} = R^{(a)}_{\text{cap}}$ in the functional form of Eq. (20), so haircuts steepen automatically as a position outgrows its exit route. Both caps follow from the same number in closed form (Eq. (12)). One dealing window clears the pool's entire debt at the point of liquidation, and the largest single borrower's debt is monitored against the same capacity number. Capacity is the named exception to the rule that missing inputs take prescribed floors, never zero (Section 4.6), because it is risk-decreasing. A missing or expired capacity input sets $R^{(a)}_{\text{cap}} = 0$, which sets both caps to zero. Controlled operational records hold the attestation schedule and launch requirements.

3.3.5 Coverage ratio

The running guardrail against aggregate exit demand is

$$\mathrm{CovR}^{(a)} = \frac{R^{(a)}_{\text{cap}}}{\sum_e B^{(a)}_e}, \tag{18}$$

aggregate borrows against one window of attested capacity, computed per asset and additionally aggregated across deployments sharing an issuer or administrator. This is the systemic check on simultaneous exit demand: demand is bounded ex ante by (12) and monitored ex post here. The exit does not run through a venue whose depth could move against it (Section 2.5). $\mathrm{CovR}^{(a)} < 1$ forces escalation level 2 (Section 5.6): new-borrow $\ell_{\max}$ cut and the supply cap stepped down, with precautionary partial redemptions to restore coverage.

3.4 Operational and wrong-way risk

3.4.1 Wrapper and transfer-agent risk

The token is a permissioned wrapper around fund shares: mint/burn authority, upgrade keys, allowlist administration, and custody of privileged roles are scored through the operational-security and custody dimensions of the due-diligence scorecard, with the residual held in the named manager/operational layer λ_{ops}. For a short-duration type the layer stack can dominate the haircut, and the named decomposition keeps that attribution visible rather than mislabelling it as SIMM-derived market risk. Where listed assets share one tokenisation platform, a shared-service-provider concentration guardrail is monitored across listed collateral sets (Section 5.5).

3.4.2 Manager and valuation-process risk

Administrator and auditor independence, the valuation verification process, restatement history, key-person provisions, and (for leveraged fund types) attested leverage versus policy are scored in the governance dimension. The governance score charges for the quality of that evidence through the due-diligence multiplier κ. The layers λ_{ops} and λ_{stale} charge for the structural failure modes per type and retain their assigned values whatever the scorecard awards. A dimension drives κ through the score or a named layer, never both. A leveraged fund type additionally takes the leverage add-on λ_{lev}, charged per turn of attested leverage above the policy level, from the administrator-signed evidence file required by the treatment.

3.4.3 Wrong-way flags

A wrong-way flag designates collateral whose stress co-realises with borrow-side stress, adding the assigned layer λ_{ww}. The clearest examples are private-credit pools whose underlying borrowers are crypto market-makers or prime brokers, whose defaults cluster with stablecoin stress, and market-neutral funds running basis strategies whose drawdowns coincide with crypto funding stress. A flag requires a documented co-realisation mechanism, and a type with none takes no flag. An intra-type coincidence between a fund's own gate and its loss event is charged in the gating layer rather than as a cross-market flag. The stablecoin peg monitor runs on the borrow side only (USD-cash policy). Peg stress that coincides with wrong-way-flagged collateral requires a more conservative intervention.

3.5 Deliberate omissions

The following metrics are omitted because they rely on observables that the methodology does not assume for collateral that exits by redemption. Price-impact slope measures and depth-based liquidity coverage require an order book or automated-market-maker depth series. Gas-breakeven and inclusion-failure probabilities require a public liquidation race, which sits outside the model's settlement assumptions. Fire-sale multipliers, price-feedback coefficients, and depth-decay coefficients require liquidation-event and market-maker-depth series, which sit outside the required public inputs. High-confidence CoVaR (co-value-at-risk) requires a tail sample, while roughly twelve appraisal-smoothed observations a year cannot identify one. The model charges for these risks through other terms. The stressed SIMM risk weights, concentration risk factor, and clearing discount reserve for the cost of clearing a position of size under stress. The prescribed SIMM correlations (ρ, γ, ψ) and shared-issuer or administrator grouping of the coverage check account for cross-asset dependence.

4 The SIMM delta aggregation engine

The engine converts the attested Common Risk Interchange Format sensitivities of Section 3 into a per-collateral haircut $H^{(a)}$, with the day-one proxies of Section 3.1.1 standing in until the first attested file arrives. The market-risk core is the ISDA SIMM v2.6 *delta* margin, inherited verbatim [1]: every risk weight RW_k, intra-bucket correlation ρ_{kl}, cross-bucket correlation γ_{bc}, and inter-class matrix $\psi_{rr'}$ is taken from the published ISDA parameter set. Nothing in this section is fitted.

The framework's own open parameters sit outside this engine. The practitioner applying the framework sets them, and each one is enumerated as a free parameter (Appendix A).

Five risk classes are in scope: Interest Rate (IR), Qualifying Credit (CreditQ), Non-Qualifying Credit (CreditNonQ), Equity, and FX. Commodity is unused. The scope is delta margin only. Section 3.1 records why a linear, long-only holding of fund units attracts neither vega nor curvature margin under SIMM ¶ 9. It also records what the framework substitutes where prepayment optionality is embedded, and why base correlation never enters.

The engine's output is an initial margin $\mathrm{IM}^{(a)}$ per unit of published valuation. The haircut then follows in closed form, Eq. (25). The pipeline is computed in one direction with no circular dependency (Section 2.5).

4.1 Weighted sensitivities and concentration

For each risk factor k with attested sensitivity s_k (DV01, CS01, or unit fund delta, expressed per unit of published valuation), the weighted sensitivity is

$$\mathrm{WS}_k = \mathrm{RW}_k\, s_k\, \mathrm{CR}_k, \tag{19}$$

with RW_k read from the v2.6 tables for the class/bucket assignment of Section 3. The concentration risk factor keeps SIMM's functional form but re-keys its threshold from dealer-market terms to the only exit route that exists for this universe:

$$\mathrm{CR} = \max\left(1, \sqrt{\frac{\text{position}}{T}}\right), \qquad T = R_{\text{cap}}, \tag{20}$$

where position is the deployment's aggregate exposure to the collateral at the current published valuation and R_{cap} is the issuer's *attested redemption capacity per dealing window.* Haircuts therefore steepen automatically, and smoothly, as a position outgrows its exit route. The same number R_{cap} drives the borrow and supply caps in Section 5. Because the supply cap satisfies $S_{\max} \le M R_{\text{cap}}$, the multiplier is bounded: $\mathrm{CR} \le \sqrt{M}$.

4.2 Within-bucket aggregation

Within each bucket b of a risk class,

$$K_b = \sqrt{\Big(\sum_{k\in b} \mathrm{WS}_k^2 + \sum_{k\in b}\sum_{l\neq k} \rho_{kl}\, f_{kl}\, \mathrm{WS}_k\, \mathrm{WS}_l\Big)_+}, \tag{21}$$

with ρ_{kl} the published intra-bucket correlation, $(x)_+ := \max(x, 0)$, and the concentration-diversification factor $f_{kl} = \min(\mathrm{CR}_k, \mathrm{CR}_l)/\max(\mathrm{CR}_k, \mathrm{CR}_l)$. f_{kl} lowers the diversification credit between factors held at different concentrations in the published standard, and under the re-keyed threshold it equals one identically. The threshold is the single number $T = R_{\text{cap}}$ and the position is the deployment's aggregate exposure to the collateral, so every factor of one collateral takes the same CR, and the same holds for g_{bc} in Eq. (23). The two factors take values below one only where the practitioner measures positions at a finer granularity than the aggregate exposure of Eq. (20). The zero floor is inert under the published correlations. FRTB applies the same guard one level up, in the across-bucket sum [2], which Eq. (22) inherits through the bounds on S_b.

4.3 Across-bucket aggregation

For each risk class r, buckets aggregate as

$$\begin{gathered}\mathrm{DeltaMargin}_r = \sqrt{\sum_b K_b^2 + \sum_b \sum_{c\neq b} \gamma_{bc}\, S_b\, S_c} \; + \; K_{\text{residual}}, \\ S_b = \max\Big(\min\Big(\sum_{k\in b} \mathrm{WS}_k,\, K_b\Big),\, -K_b\Big).\end{gathered} \tag{22}$$

The bucket sums S_b, bounded to $[-K_b, K_b]$, keep each bucket's signed contribution within its own stand-alone margin, so cross-bucket terms can never manufacture risk a bucket does not hold. The Residual bucket is *additive* (it receives no diversification benefit), which is precisely the mechanism that charges for a missing sector or quality attestation (Section 4.6).

4.4 The interest-rate variant

The IR class differs in three inherited respects. First, buckets are currencies, and the published standard applies concentration *per currency*. The framework instead evaluates CR on the asset's aggregate exposure at the published valuation, exactly as Eq. (20) does. The re-keyed threshold $T = R_{\text{cap}}$ is a capacity in USD, so the position compared against it is an amount in USD rather than a sum of sensitivities. A per-currency multiplier arises only where the practitioner measures the position by currency, and CR_b then reads the USD of published valuation whose interest-rate sensitivities sit in currency b. The within-currency aggregation uses the single CR_b rather than per-factor f_{kl}. Second, distinct sub-curves of the same currency correlate at the published sub-curve parameter $\rho_{\text{sub}} = 0.993$, applied multiplicatively to the tenor correlation ρ_{kl} in Eq. (21). A portfolio in this universe typically holds the overnight indexed swap (OIS) sub-curve only, so ρ_{sub} enters only where a second sub-curve is held, for example a USD *Prime* sub-curve on floating-rate consumer-credit collateral. Third, the cross-currency term scales by the concentration ratio:

$$\text{DeltaMargin}_{\text{IR}} = \sqrt{\sum_b K_b^2 + \sum_b \sum_{c \neq b} \gamma_{bc}\, g_{bc}\, S_b\, S_c}, \qquad g_{bc} = \frac{\min(\text{CR}_b, \text{CR}_c)}{\max(\text{CR}_b, \text{CR}_c)}. \tag{23}$$

For the predominantly single-currency (USD) portfolios in this universe, Eq. (23) reduces to K_{USD}.

4.5 Risk-class roll-up and the haircut

Writing $\text{IM}_r := \text{DeltaMargin}_r$, the risk classes a SIMM product class p holds combine through the published inter-class matrix $\psi_{rr'}$, and the product-class margins then add:

$$\text{IM}_p = \sqrt{\sum_{r \in p} \text{IM}_r^2 + \sum_{r \in p} \sum_{r' \neq r} \psi_{rr'}\, \text{IM}_r\, \text{IM}_{r'}}, \qquad \text{IM} = \sum_p \text{IM}_p. \tag{24}$$

This is the strict SIMM v2.6 reading: the product-class margins, RatesFX and Credit among them, take no diversification between them. Where each product class holds one risk class, as it does for the collateral types in scope, that addition reduces to the plain sum of the class margins and $\psi_{rr'}$ never enters. Margining the collateral as a single product class instead would diversify its rate and credit legs under $\psi_{rr'}$, and since $\psi_{rr'} \leq 1$ it would return the smaller number. The two readings differ in aggregation structure rather than in any published value.

SIMM is calibrated to a 99% confidence level over ten business days, and $\text{IM}^{(a)}$ enters at that horizon *verbatim*, with no rescaling for the margin period of risk. Those ten business days are the reference horizon of the inherited standard. Section 2.2 states the attribution convention that splits the reserve at ten business days. The extension to the full clearing window is charged through a separate haircut component, δ_{clear}, defined next. Keeping IM at the native ten-business-day horizon is deliberate, because exact reconciliation against the published ISDA standard holds only if the market-risk core is the unmodified standard.

The clearing discount δ_{clear}. The clearing discount reconciles the SIMM charge at ten business days to the price risk of the governing clearing window, as an additive haircut component held *above* the liquidation threshold. Its single defining equation is Eq. (6) (Section 2), which fixes the day-one value, the terms a margin credit requires, and the floor the rescaled move sets on the total price-risk charge. The framework does not prescribe the clearing mechanism.

The haircut is

$$H = \mathrm{IM} + \delta_{\text{clear}} + \delta_{\text{carry}} + \Lambda, \qquad \Lambda = \sum_j \lambda_j, \tag{25}$$

where Λ is the sum of the additive layers: gating and suspension, valuation staleness and smoothing, oracle pipeline integrity (0 at the robust configuration), a default add-on ($PD \times LGD$) for private credit, prepayment for amortising collateral, manager and operational, wrong-way, a leverage add-on, and an event-risk layer for event-linked collateral (the ninth named layer slot). Each layer is expressed as a percentage per listing. The practitioner sets the model constants used to determine it. Together they charge for exactly what a sensitivity-based margin structurally cannot see (narrowed redemptions, smoothed valuations, jump-to-default, wrapper and wrong-way risk). The interest cost δ_{carry} is the certain cost of holding the debt over the clearing window, defined once at Eq. (7) of Section 2. Policy limits follow immediately, $\theta^{(a)} = \kappa^{(a)}\big(1 - H^{(a)}\big)$ and $\ell^{(a)}_{\max} = \theta^{(a)} - \nu$, with $\theta^{(a)} = 1 - H^{(a)}$ at the Prime tier. The static liquidation buffer ν is a per-listing constant within the 4–8 percentage-point range. Section 5 specifies the public constraints on that constant, the caps, and the due-diligence tier multipliers.

4.6 Missing inputs take prescribed floors

If any required input is unavailable, it is replaced by a *prescribed conservative floor* in the spirit of FRTB's non-modellable risk factor treatment, never by zero. Concretely: a missing sector or quality attestation routes the exposure to the class's Residual bucket (highest risk weight, additive aggregation per Eq. (22)). A missing fund-unit attestation file reclassifies the position to Equity Residual, whose published risk weight puts $H^{(a)}$ above one at any window from forty business days (Section 4.7).

The clearing window LH of Eq. (4) takes no floor. The three-source hierarchy of Section 2.1 always names a governing window: with neither an attested service-level agreement nor a horizon stated in the offering documents, the regulatory fallback governs and the worst-case wait is added to it. Observed redemptions are never averaged into the window at any level of the hierarchy.

An unattested redemption capacity leaves the threshold T of Eq. (20) with no admissible value. The pre-signature document proxy supplies the conservative default from the offering document's stated per-cycle gate fraction g times assets under management, $R_{\text{cap}} = g \cdot \mathrm{AUM}$, for the first ramp step. Further exposure requires a signed attestation, which supersedes that default. An uncured lapse or refusal sets $R_{\text{cap}} = 0$ instead, which sets both caps to zero and stops new business against the asset.

4.7 Boundedness and the eligibility rule

H is bounded at each recomputation, component by component. The sensitivities s_k, per unit of published valuation, are bounded by the asset's duration, weighted-average life, or unit delta. The RW_k are fixed published magnitudes. $\mathrm{CR} \le \sqrt{M}$ holds under the supply cap (Section 4.1). The model constants set by the practitioner for each layer remain fixed between reviews, so the baseline part of Λ is a finite sum of constants at each recomputation.

The day-one clearing discount is bounded on both sides by the windows that can govern the asset. It equals $\mathrm{IM}\big(\sqrt{\mathrm{LH}/10} - 1\big)$ (Eq. (6)), which is largest at the longest admissible LH. Under the regulatory fallback of the hierarchy, that window is the FRTB securitisation row of 120 business days plus the worst-case wait to the next submission deadline. An asset whose documented contractual window exceeds the FRTB row takes that contractual window instead, again with the wait added. Either way the governing window is documented, so the value is a finite number known at each recomputation. The negative side is bounded by the policy minimum of one business day, so $\mathrm{IM} + \delta_{\mathrm{clear}}$ never sits below $\mathrm{IM}\sqrt{1/10}$ and the credit never exceeds 0.684 IM.

Three inputs can rise above those bounds while an asset stays listed. Qualified clearing observations enter δ_{clear} as their conservative quantile (Eq. (6)), so an observed clearing cost above the rescaled move governs the total price-risk charge, and no model parameter caps it. A widening of the realised clearing window under sustained settlement failure (Section 5.6) raises the rescaled move itself. The staleness escalation $\Delta\lambda_{\mathrm{stale}}$ added to the baseline set by the practitioner grows with the time a publication is overdue (Eq. (13)). Each of the three drives H toward the cutoff at one rather than toward a ceiling below it. The practitioner-facing rule catches the rise:

> *If the computed $H^{(a)} \geq 1$, the collateral is ineligible. $H \geq 1$ is the model stating that one clearing window of risk exceeds the position's value. It is never interpreted as a negative LTV.*

Non-attestation is the case the rule is built for. A market-neutral fund that fails to deliver its administrator attestation file reclassifies from Equity bucket 11 to Equity Residual, at the published SIMM risk weight for that bucket (50 at v2.6). At an illustrative 45-business-day window, the $\sqrt{\mathrm{LH}/10}$ extension takes $\mathrm{IM} + \delta_{\mathrm{clear}}$ alone above one before any layer is added, so the asset is ineligible by the rule above. That is the intended consequence of withholding the file.

4.8 A treasury money-market fund worked end to end

The example runs one asset through the whole engine, from a fact sheet to the synthetic limits. The asset is a tokenised government money-market fund whose administrator publishes a daily net asset value (NAV, the published valuation for this asset), with a reported weighted-average maturity of 32 days. Its fact sheet gives the maturity distribution: 35% of NAV maturing within 14 days, 35% between 15 and 30 days, 25% between 31 and 90 days, and 5% between 91 and 180 days. Every risk weight and correlation below is the published v2.6 constant [1], and the sensitivities are day-one estimates built from the fact sheet. An administrator-attested Common Risk Interchange Format file replaces them when delivered.

Bucket assignment starts the chain. Each maturity range goes to the nearest longer SIMM vertex at that vertex's own tenor, so the round-up works in the framework's favour. The DV01 at each vertex is the range weight, as a fraction of NAV, times the vertex tenor. The maturity ladder in Table 1 sums to 0.1301 bp of NAV per bp, a rate duration of 47.5 days against the reported 32. The residual sovereign-issuer spread exposure is recorded as the same 0.1301 bp of CS01 in CreditQ bucket 1 (investment-grade sovereigns, RW = 75) under the day-one single-name treatment of Section 3.1.1. The position sits inside the attested redemption capacity, so every concentration risk factor is one (CR = 1, hence $f_{kl} = 1$), and only the overnight indexed swap sub-curve is held, so ρ_{sub} does not enter.

Table 1 reports each input per unit of published valuation. Maturity weights are percentages of published valuation, and vertex tenors are in years. DV01 s_k is in basis points of published valuation per basis-point move. The weighted sensitivity $\mathrm{WS}_k = \mathrm{RW}_k s_k/100$ is a percentage of published valuation at $\mathrm{CR}_k = 1$ (Eq. (19)). The risk weights are in basis points and come from the published v2.6 tables [1].

The IR class is a single currency bucket, so Eq. (21) applies once, with the published short-end tenor correlations ($\rho_{2w,1m} = 0.77$, $\rho_{2w,3m} = 0.67$, $\rho_{2w,6m} = 0.59$, $\rho_{1m,3m} = 0.84$, $\rho_{1m,6m} = 0.74$, $\rho_{3m,6m} = 0.88$):

$$\mathrm{DeltaMargin}_{\mathrm{IR}} = K_{\mathrm{USD}} = \sqrt{\underbrace{0.004631}_{\sum_k \mathrm{WS}_k^2} + \underbrace{0.007556}_{\sum_{k\neq l} \rho_{kl}\mathrm{WS}_k\mathrm{WS}_l}} = 0.1104\%. \tag{26}$$

Table 1: Day-one input estimates and weighted sensitivities for the worked example.

USD IR vertex	2w	1m	3m	6m
Maturity weight	35	35	25	5
Vertex tenor	0.0384	0.0833	0.2500	0.5000
DV01 s_k	0.0134	0.0292	0.0625	0.0250
RW_k	109	105	90	71
WS_k	0.0146	0.0306	0.0563	0.0178

The credit leg is one risk factor in one bucket, $\mathrm{WS}_{\mathrm{CQ}} = 75 \times 0.1301/100 = 0.0976\%$ of NAV, so Eq. (22) returns it unchanged as the class margin. The rate leg sits in the RatesFX product class and the credit leg in Credit, each holding one risk class, so the two margins add and $\psi_{rr'}$ has no pair to diversify:

$$\mathrm{IM} = K_{\mathrm{USD}} + \mathrm{WS}_{\mathrm{CQ}} = 0.1104\% + 0.0976\% = 0.2080\% \tag{27}$$

per unit NAV, the unmodified SIMM charge at ten business days. Every number in the chain traces in two steps to a published v2.6 table entry or to a fact-sheet line.

The policy outputs follow in one pass. The fund deals daily, and its attested terms give the clearing window of Eq. (4) as three business days. The legs are one day of worst-case wait to the next dealing deadline, no notice period, one dealing day, and settlement on the following business day. Eq. (6) then returns

$$\delta_{\mathrm{clear}} = \mathrm{IM}\left(\sqrt{\tfrac{3}{10}} - 1\right) = -0.09\%, \tag{28}$$

a margin credit, because the position reaches cash inside SIMM's ten-business-day horizon. The synthetic example assumes no qualified clearing observation that would raise the charge.

The remaining inputs are the practitioner's own elections, shown here at illustrative values. The named layers total $\Lambda = 2.0\%$ (gating and suspension 0.5, valuation staleness and smoothing 0.5, manager and operational 1.0). The ceiling borrow rate is $r^{B,\max} = 10\%$, the market's annual percentage yield at full utilisation, the due-diligence tier is the top one ($\kappa = 1$), and the liquidation buffer is $\nu = 4\,\mathrm{pp}$. Then $c = r^{B,\max}\,\mathrm{LH}/252 = 0.001190$, i.e. 0.1190%, and Eq. (8) gives $H = (0.1139 + 2.0 + 0.1190)\%/(1.001190) = 2.23\%$ with $\delta_{\mathrm{carry}} = 0.12\%$, so Eq. (25) reads

$$\begin{aligned} &H = \mathrm{IM} + \delta_{\mathrm{clear}} + \delta_{\mathrm{carry}} + \Lambda = 0.2080\% - 0.0941\% + 0.1164\% + 2.0000\% = 2.2303\%, \\ &\theta = 1 - H = 97.7697\%, \end{aligned} \tag{29}$$

and the operating LTV ceiling is $\ell_{\max} = \theta - \nu = 93.7697\%$. The composition is the point of the example: the illustrative layers contribute 2.0 of the 2.23, so the synthetic limits reflect the named layers far more than the inherited SIMM core. The three-day window is worth 0.36 pp of borrow limit against the same fund at a window of ten business days.

The same fund at monthly dealing shows the clearing window doing work. Take a monthly dealing day with orders due five business days before it and cash paid five business days after. An order that just misses a deadline waits up to 22 business days for the next dealing day, a rounded worst case for a monthly cycle. Notice, the dealing day, and settlement follow, so $\mathrm{LH} = 33$ business days. The clearing discount of Eq. (6) turns from a credit into a charge, $\delta_{\mathrm{clear}} = 0.2080\% \times (\sqrt{33/10} - 1) = +0.17\%$, the interest cost grows with the window to $\delta_{\mathrm{carry}} = 1.26\%$, and the same elections raise the haircut from 2.23% to $H = 3.64\%$, with $\ell_{\max} = 92.36\%$. Nothing else about the asset changed: the increase is the price of waiting more than ten times as long for the issuer to settle, and an attested faster window would earn it back.

5 From haircut to lending limits

The four outputs of Section 1 follow in closed form, in one pass with no circular dependency and no iteration. The liquidation threshold and the borrow limit come from the aggregation engine of Section 4, through the recipe of Eq. (1). The two caps come from attested capacity, below. The policy layer adds five elements to the recipe. Three act on the operating range: the static liquidation buffer $\nu^{(a)}$ (Section 5.1), the due-diligence tier multiplier $\kappa^{(a)}$ (Section 5.2), and the qualifying-insurance credit against the named operational layers (Section 5.3). Two act on quantity and on state: the borrow and supply caps with their launch ramp (Section 5.4), and the valuation-native guardrails with their hard overrides (Section 5.5).

The clearing discount $\delta^{(a)}_{\text{clear}}$ takes its single defining equation from Eq. (6), evaluated on the clearing window $\text{LH}^{(a)}$ that the three-source hierarchy of Section 2.1 fixes. It is an additive haircut component *inside* H, above the liquidation threshold. A window beyond ten business days makes it a charge. Contractually enforceable terms, attested or documented, that enforce a shorter window make it a credit. Estimated legs and FRTB fallback windows never earn one.

Caps follow from the issuer's attested redemption capacity per dealing window $R^{(a)}_{\text{cap}}$ (Section 5.4). Capacity is the named exception to the floors of Section 4.6. Documented capacity evidence is a listing precondition rather than an input that takes a floor. The offering-memorandum proxy supports the first ramp step. Further exposure requires a signed attestation. An uncured lapse or refusal sets $R^{(a)}_{\text{cap}} = 0$, which sets both caps to zero.

The split of the price-risk charge into $\text{IM}^{(a)}$ and $\delta^{(a)}_{\text{clear}}$ leaves the reserve sufficient at the point of liquidation and costs nothing numerically. The argument for that sufficiency and the three reasons for the split are stated once, in Section 2.

5.1 Static liquidation buffer

The liquidation buffer $\nu^{(a)}$ is fixed rather than recomputed from live conditions. It is a per-listing constant, the gap between $\theta^{(a)}$ and $\ell^{(a)}_{\max}$, inside the 4–8pp range of Section 2.4. The reference check of Eq. (11) tests the adopted constant and does not recompute it from live data. A listing whose implied minimum exceeds its adopted buffer launches with a tighter ceiling for new borrows only, through the guardrails of Section 5.5, rather than with a wider buffer. Assets with sparse published valuations, or with a documented exposure that a single adverse valuation could realise as a jump, are the listings that need the tighter ceiling. A state-dependent gap is procyclical: it shrinks the buffer between live positions and the liquidation threshold exactly when published valuations fall and the redemption process is slowest, pushing performing positions into liquidation at the worst point of the cycle. Stress response instead uses those same guardrails, which can only tighten. None of them retroactively liquidates a performing position.

5.2 Due-diligence tier multipliers

Eligibility and final limits pass through the due-diligence ranking framework (eligibility failures first, then the weighted ordinal score over seven dimensions, with the full rubric in the practitioner's scorecard). The dimensions are legal structure, redemption firmness, regulatory status, transparency, operational security, custody, and governance. An asset with a bridge or cross-chain dependence sits outside the scope of this framework and is not evaluated under it. That covers a bridged or wrapped token, and mint/burn, transfer agency, or a valuation oracle depending on cross-chain messaging. No score rescues an eligibility failure. For assets that pass, the tier multiplier $\kappa^{(a)} \in (0,1]$ scales the type's threshold, $\theta^{(a)} = \kappa^{(a)}\big(1 - H^{(a)}\big)$, and the borrow LTV follows as $\ell^{(a)}_{\max} = \theta^{(a)} - \nu^{(a)}$, with the liquidation buffer $\nu^{(a)}$ absolute rather than tier-scaled. The over-collateralisation cushion at the point of liquidation is therefore $1 - \theta^{(a)} \geq H^{(a)}$, strict below the Prime tier.

The tier structure, the thresholds, and the multiplier per tier are the practitioner's own elections, fixed in their scorecard (free parameter, Appendix A). An illustrative structure has three tiers and a reject decision. Prime is a score of at least 80 with a minimum of 4 on legal structure, redemption firmness, and transparency, nothing below 3 on any dimension, and $\kappa = 1.00$. Standard is a score of at least 65 with nothing below 2 on any dimension and $\kappa = 0.90$. Restricted is a score of at least 50 with $\kappa = 0.75$ and a concentration penalty the practitioner elects (free parameter, Appendix A). Reject is a score below 50, any eligibility failure, or a zero on any dimension.

The tier acts on the over-collateralisation cushion only, and the layer table $\Lambda^{(a)}$ is tier-invariant. A scorecard dimension drives $\kappa^{(a)}$ through the weighted score or it drives a named layer, never both: each fact is charged exactly once. Transparency and governance drive $\kappa^{(a)}$. The effective oracle tier sets λ_{oracle} and caps the transparency score.

5.3 Qualifying-insurance credit

Genuine, enforceable insurance reduces the charge. It can reduce *only* the named operational/process layers it demonstrably covers, never the market or liquidation reserve. For a qualifying policy covering layer $\lambda_j^{(a)}$,

$$\lambda_j^{(a),\text{eff}} = \max\Big(\phi_j\,\lambda_j^{(a)},\ \ \lambda_j^{(a)} - R_{\text{ins},j}^{(a)}\Big), \qquad R_{\text{ins},j}^{(a)} = \beta_j\,\lambda_j^{(a)}\ \min\Big(1, \tfrac{\text{NetLimit}}{E_{\text{ref}}^{(a)}}\Big), \tag{30}$$

where $E_{\text{ref}}^{(a)} = \max(\text{live exposure}, S_{\max}^{(a)})$, NetLimit is the per-occurrence limit *net* of deductibles, retentions, shared-limit allocation, and excluded perils, $\beta_j \in [0,1]$ is the prescribed recognition factor for the policy's evidenced quality, and ϕ_j is a prescribed floor. The recognition factors, floors, rating minimum, and aggregate cap named below are practitioner elections registered in Appendix A. The credit is bounded as follows:

(i) **Eligible layers only:** the manager/operational layer λ_{ops} (custody, crime, cyber, or operational cover) and the oracle-integrity layer λ_{oracle} (professional-indemnity or valuation-error cover). It *never* reduces the price-risk charge $\text{IM}+\delta_{\text{clear}}$, the valuation-staleness layer (baseline or its dynamic overdue-publication escalation), or the gating, default, wrong-way, or leverage layers.

(ii) **Capped and floored:** the reduction never exceeds the layer ($\beta_j \le 1$) and never takes it below $\phi_j\lambda_j$, where the practitioner sets the floor fraction ϕ_j. One policy offsets exactly one layer (no stacking). The total reduction across all layers is bounded after the per-layer credits by an aggregate cap the practitioner sets. Where that aggregate cap governs, the reduction is allocated to the worst layer first.

(iii) **Qualification (else $\beta_j = 0$):** a regulated insurer rated at or above the minimum the practitioner sets, in-force with premium paid, scope matched to a *modelled* loss, for-benefit-of-lenders or loss-absorbing seniority, adequate limit, exclusions reviewed to confirm the policy pays in the modelled scenario, and a claims-paying record. Affiliated/captive or balance-sheet "self-insurance" scores $\beta_j = 0$. Insurance never cures an eligibility failure (single-key control, sanctions, fraud, unrecovered loss) and never brings a bridge-dependent asset into scope.

(iv) **Recognition tiers**, set by the practitioner: standard qualifying, strong (broad wording, clean claims), and cash-collateralised / AA-insurer / letter-of-credit-equivalent. Grades are capped at $1 - \phi_j$, the largest recognition the floor admits.

(v) **Lapse:** non-payment, cancellation, rating downgrade below the rating minimum, limit exhaustion, or a relevant-peril exclusion withdraws the reduction immediately for new business ($R_{\text{ins}} \to 0$). The reduction returns only when the policy again qualifies.

Until the practitioner sets β_j, ϕ_j, and the rating minimum, $\beta_j = 0$ and no reduction is granted. The graduated insurance evidence also feeds the legal-structure and operational-security dimensions of the due-diligence scorecard, replacing the binary top-level insurance condition. To prevent a double benefit, a policy recognised here for a layer reduction ($R_{\text{ins}} > 0$) supports the levels awarded on those dimensions only as disclosure and enforceability evidence, and it does *not* additionally raise the tier multiplier κ for the same coverage.

5.4 Caps and launch ramp

Both caps are keyed to the attested issuer per-window redemption capacity $R^{(a)}_{\text{cap}}$. The same number re-keys the SIMM concentration threshold T inside the concentration risk factor $\text{CR}^{(a)} = \max\{1, \sqrt{\text{position}^{(a)}/R^{(a)}_{\text{cap}}}\}$ of Section 4:

$$B^{(a)}_{\max} \;\leq\; R^{(a)}_{\text{cap}}, \qquad S^{(a)}_{\max} \;\leq\; M\, R^{(a)}_{\text{cap}}, \tag{31}$$

with the policy constant M, the supply-cap multiple, a free parameter of the practitioner (Appendix A). At the illustrative election $M = 8$, $S^{(a)}_{\max} \leq 8\, R^{(a)}_{\text{cap}}$ and the re-keyed concentration risk factor is bounded at $\text{CR} \leq \sqrt{M} \approx 2.83$. The borrow cap is further bounded by the committed settlement level of Eq. (35), drawn from the subscription documents (Section 6). $B^{(a)}_{\max}$ bounds borrowed *debt* against collateral a as a pool-level parameter of the deployment. The deployed market configuration shows the active borrow and supply caps. The practitioner sets M as a model constant. $R^{(a)}_{\text{cap}}$ and $S^{(a)}_{\max}$ are in the collateral's published-valuation USD numéraire.

Per-issuer global aggregation. Eligibility, tiering, and capacity are judged once per issuer rather than per deployment. Where the same issuer's token is listed in multiple deployments, a single global issuer cap governs the sum of per-deployment caps,

$$\sum_{\text{deployments}} S^{(a)}_{\max} \;\leq\; S^{(\text{issuer})}_{\text{global}} \;\leq\; M\, R_{\text{cap}}, \tag{32}$$

because R_{cap} is an issuer-level resource that does not multiply with the number of listings. A policy consequence triggered at issuer level applies across every deployment that references the issuer.

The liquidator redeems collateral at the published valuation only until the debt is repaid. One dealing window must therefore clear the largest position's *debt* at the point of liquidation, $B^{(a)}_{\max} \leq R^{(a)}_{\text{cap}}$, and excess collateral above the repaid debt queues at the borrower's risk (Section 6). $R^{(a)}_{\text{cap}}$ enters here on the one-window basis of Section 2.5. A capacity attested on another basis is normalised to the governing dealing window.

During launch, the active on-chain cap may sit below Eq. (31). The controlled operations record holds the launch schedule. Restricted-tier assets take the concentration penalty of Section 5.2.

5.5 Valuation-native guardrails and hard overrides

Guardrails read observations from the redemption process itself: valuation publication, gates, capacity, and ratings. Every guardrail is automatic and can only tighten:

Valuation not published within the contractual SLA.
Pause new borrows ($\ell_{\max} \to 0$ for new positions). Existing positions are untouched. Recurrent breaches escalate the staleness layer.

Gate, suspension, or holdback announced by the issuer.
Pause the asset (no new borrows or collateral inflow) and start an on-chain cap reduction toward orderly full redemption. θ on live positions is never lowered (Section 1.3). A disorderly or unresolved redemption gate bars relisting. A resolved gate remains subject to controlled revalidation.

Coverage-ratio breach (aggregate borrows exceed one $R^{(a)}_{\text{cap}}$ window).
Cut $B^{(a)}_{\max}$ and $S^{(a)}_{\max}$.

Securitised-collateral rating falls below investment grade.
New borrows pause and caps step down immediately. *Confirmed* loss of investment grade reassigns new business to CreditNonQ bucket 2 and starts automatic full redemption. That bucket's published SIMM risk weight (1,300 at v2.6) raises $H^{(a)}$ toward the eligibility cutoff of Section 4.7.

Proof-of-reserves attestation failure or valuation restatement.
The failure can block relisting or tighten the due-diligence score.

The numeric alert levels behind the guardrails above, and the escalation level each action belongs to, are specified in the operations companion (Section 5.6). The stablecoin peg on the borrow leg carries no charge under the USD-cash borrow policy. Hard overrides take precedence over the model: when any of the conditions above is met, its action applies regardless of what Eq. (1) or the tier multipliers would otherwise permit. Restoration to model-implied parameters occurs only after the guardrail condition clears.

5.6 Operating rules and escalation

Automatic actions respond to adverse evidence and can only tighten active limits or pause new exposure. They never lower the liquidation threshold on a live position. An active value can loosen only after the condition that tightened it has cleared. The operations companion specifies the alert levels, monitoring, retention, procedures, and approvals.

6 Redemption execution and feasibility

A position in liquidation exits only through the issuer's own subscription and redemption process at the published valuation. Any on-chain secondary market is out of scope for this version, and an asset with a bridge or cross-chain dependence sits outside the framework's scope altogether (Section 1.1). The position is warehoused through the clearing window instead: it is held from the breach to cash settlement, over the clearing window $\mathrm{LH}^{(a)}$ of Eq. (4). The framework reserves the discount that compensates whoever clears the position, and it does not prescribe the clearing mechanism. The holder is the deployment or an entity it designates rather than an outside counterparty standing ready at the moment of need (Appendix A).

The discount is bounded by the haircut less its layer total. Liquidation begins when a position's loan-to-value ratio reaches $\theta^{(a)}$. The over-collateralisation cushion above the liquidation threshold is therefore $1-\theta^{(a)} \geq H^{(a)} = \mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \delta^{(a)}_{\text{carry}} + \Lambda^{(a)}$ (equality at the Prime tier), and whatever discount compensates the clearing party is paid out of it:

$$\text{discount} \;\leq\; \mathrm{IM}^{(a)} + \delta^{(a)}_{\text{clear}} + \delta^{(a)}_{\text{carry}} \;=\; H^{(a)} - \Lambda^{(a)} \;\leq\; H^{(a)}. \tag{33}$$

The bound is non-negative at every admissible window (Section 2.4). Whether a party clears at that reserved level is a stated assumption of the framework (assumption A17 of the assumptions addendum). The clearing discount is reserved rather than fitted to the valuation history.

6.1 Redemption capacity

6.1.1 Issuer capacity and committed settlement

The redemption capacity of collateral a over one dealing window is

$$Q^{(a)}_{\text{cap,red}} \;=\; \min\Big\{\,Q^{(a)}_{\text{issuer}},\; Q^{(a)}_{\text{queue}}\,\Big\}, \qquad Q^{(a)}_{\text{issuer}} \;=\; R^{(a)}_{\text{cap}}, \tag{34}$$

where $R^{(a)}_{\text{cap}}$ is the issuer's attested per-window redemption capacity (USD), obtained from the issuer or transfer agent, used as the one-window capacity consistently with the caps, the concentration threshold T, and the coverage ratio (Section 2.5). The normalisation of attested capacity onto the dealing window is part of the definition of $R^{(a)}_{\text{cap}}$ there. For an evergreen fund dealing monthly under a hard quarterly redemption gate the question inverts structurally. The gate rather than the dealing calendar sets what one window can clear, so the gate window is the governing dealing window, and the window-normalisation rule of Section 2.5 is met on that basis. $Q^{(a)}_{\text{queue}}$ is specified from the issuer's own documents, the subscription documents together with the offering memorandum's gate and dealing provisions, rather than estimated from market data.

6.1.2 The two endpoints of queue capacity

Feasibility turns on the worst credible recovery in a saturated window, and that recovery has exactly two endpoints.[2]

$$Q^{(a)}_{\text{queue}} \;=\; \begin{cases} 0 & \text{the subscription documents commit to no measurable within-window processing level, so within-window processing is not guaranteed and the conservative saturated-window floor applies,} \\ \geq\; q_L^{\text{committed}} & \text{a documented or attested committed settlement level guaranteeing processing within the window.} \end{cases} \tag{35}$$

The committed-settlement term defaults to the within-window processing level committed in the issuer's subscription documents. That level is read case by case per listing, since those terms vary from one issuer to the next. A higher attested committed settlement level is recognised where one exists.

The borrow cap is zero on the first branch of Eq. (35), so the asset can be listed and cannot be borrowed against, and the launch ramp of Section 5.4 applies only once the committed-settlement term is positive. On the second branch, $Q^{(a)}_{\text{queue}}$ is bounded below by the committed settlement level $q_L^{\text{committed}}$, and the check passes only where that level covers the borrow cap. The same evidence shortens the clearing window $\mathrm{LH}^{(a)}$ of Eq. (4), and neither reduction is a condition of listing. Documented pro-rata terms do not by themselves establish a guaranteed within-window clear. They feed the stressed realised $\mathrm{LH}^{(a)}$ and the clearing-discount reserve (Eq. (6), Section 3.3) rather than a positive committed-settlement term.

[2]Full derivation. Let $g^{(a)}$ be the documented gate fraction per dealing window, $A^{(a)}$ the fund AUM, q_L the redemption request submitted for the position in liquidation, and q_O^* the stressed concurrent third-party demand. The window fills $\min\{q_L + q_O^*,\, g^{(a)}A^{(a)}\}$, and under documented allocation the recovery on q_L is $Q^{(a)}_{\text{queue}} = q_L \cdot \min\{1,\, g^{(a)}A^{(a)}/(q_L + q_O^*)\}$ (pro-rata) or $Q^{(a)}_{\text{queue}} = \min\{q_L,\, \max(0,\, g^{(a)}A^{(a)} - q_O^*)\}$ (first-come-first-served, with q_L at the back of the queue). The stress input q_O^* is prescribed: the largest attested historical window demand where the transfer agent reports queue depth and gate utilisation, else the conservative floor $q_O^* = g^{(a)}A^{(a)}$ (a fully subscribed window) where unattested, consistent with the missing-data rule of Section 4.6.

6.1.3 Clearing window per collateral

The clearing window $\mathrm{LH}^{(a)}$ over which the position is warehoused is the window of Eq. (4), fixed by the three-source hierarchy of Section 2.1. It is the same quantity on which $\delta^{(a)}_{\text{clear}}$ is built (Section 2).

An attested LH, and the δ_{clear} and liquidation threshold derived from it, hold only while the attested terms behind the governing window remain in force. When they lapse, the next source in the hierarchy of Eq. (4) governs and the limits of Section 5 re-derive accordingly.

6.2 Feasibility condition

The sole capacity check of the framework is that the largest permissible debt position exits within one dealing window:

$$B^{(a)}_{\max} \;\le\; Q^{(a)}_{\text{cap,red}} \qquad \text{over } \mathrm{LH}^{(a)}, \tag{36}$$

with $q_L = B^{(a)}_{\max}$ in Eq. (35). The prescribed stress is the pool's entire debt at the borrow cap, measured at the point of liquidation, clearing one window with the whole position redeemed. The interest accruing over that window is reserved separately by $\delta^{(a)}_{\text{carry}}$. Only the debt amount must clear the window, since the proceeds of the issuer redemption are applied to the debt first. Any collateral above the repaid debt queues into the following window at the borrower's risk rather than the suppliers'. Since Section 5 sets $B^{(a)}_{\max} \le R^{(a)}_{\text{cap}}$ by construction, the issuer capacity term of Eq. (34) holds automatically. The operative content of Eq. (36) is the committed-settlement term, which by Eq. (35) reduces to the documented committed settlement level.

The check is computed in one pass, in one direction: the caps from Section 5 feed in, Eq. (36) either passes or fails, with no iteration back into the haircut. The check uses the current attested inputs. If the condition fails, limits must tighten: $B^{(a)}_{\max}$ and $S^{(a)}_{\max}$ are reduced until Eq. (36) passes. Where debt can be attributed to entities, the horizon for the largest single debt is $\lceil B^{(a)}_{(1)}/R^{(a)}_{\text{cap}} \rceil$ windows, and under the supply cap alone a full redemption completes within M windows (Section 2.5).

6.3 Settlement evidence and model consequences

Every parameter in this section is set on the floor state: no measurable committed settlement level in the subscription documents, no clearing history, and no party standing ready at the moment of need. Qualifying settlement evidence earns a practitioner-set reduction against that floor (free parameter, Appendix A). The model recognises three forms of evidence and one live control:

(i) **The committed settlement level.** The subscription documents of a listed collateral give the processing level committed within the window.

(ii) **Test redemptions.** A completed test corroborates the attested $\mathrm{LH}^{(a)}$. A failed test, or one that settles beyond the attested window, widens it to the documented worst case. The operations companion specifies the test procedure.

(iii) **Settlement records.** A realised window longer than the attested one widens $\mathrm{LH}^{(a)}$ and raises the clearing-discount reserve of Eq. (6). That record can only tighten the parameters.

(iv) **Rate limit on queue saturation.** Queue saturation or issuer processing delays trigger a rate limit on new borrows. The operations companion specifies the release procedure (Section 5.6).

6.3.1 Redemption timing evidence

Redemption evidence compares the realised time from notice to settlement with the attested window. An overrun widens $\mathrm{LH}^{(a)}$ under item (iii) above. The operations companion specifies the notice, reconciliation, and recordkeeping procedures (Section 5.6).

Appendix A: Free parameters

This appendix lists the open model quantities that enter the calculation or constrain its on-chain outputs. A model constant is a value that the practitioner sets outside the calculation and supplies to it as a fixed input. Active LTV, LLTV, and cap values are visible in the deployed market configuration. Operational and validation records remain controlled.

Every open model quantity follows the constraints stated here. A practitioner may tighten an active on-chain limit at any time. A loosening requires a documented decision on current evidence. The overdue-publication adjustment in Eq. (13) reverses automatically when its public input recovers, as the published equation requires.

Clearing window LH. The source hierarchy is fixed by Section 2.1. The practitioner sets a fallback horizon where the cited standard provides no natural horizon for the class. The clearing window satisfies $\mathrm{LH} \geq 1$ business day, and the FRTB fallback constants remain those of the cited standard [2]. Less certain settlement terms justify a longer window.

Clearing discount δ_{clear}. The value is fixed at $\mathrm{IM}\big(\sqrt{\mathrm{LH}/10} - 1\big)$, a charge for a window beyond ten business days and a margin credit for a shorter one. Only contractually enforceable legs support a margin credit. The controlled validation record fixes N_{obs}, the observation quantile, the look-back period, and the diversity requirement. The total price-risk charge $\mathrm{IM} + \delta_{\mathrm{clear}}$ remains at least $\mathrm{IM}\sqrt{\mathrm{LH}/10}$.

Additive layers λ_j. The layer slots are fixed by Section 2.3. The practitioner sets each non-negative model constant. Each derived layer takes its inputs from its stated source. The cited public standards supply each prescribed value.

Tier multiplier κ. The multiplier for each due-diligence tier is a model constant set by the practitioner. Any concentration penalty attached to a tier is also a model constant set by the practitioner. The multiplier satisfies $\kappa \in (0, 1]$, with the top tier at 1. The model assigns each due-diligence dimension to exactly one of κ and a named layer.

Liquidation buffer ν. The practitioner sets ν as a model constant for each listing. The stressed-jump floor $q_p(J)$ and its quantile level p are model constants set by the practitioner. The liquidation buffer remains between 4 and 8 percentage points and must satisfy Eq. (11).

Capacity and caps. The practitioner sets the supply-cap multiple M. It is an integer of at least 1 and bounds the concentration risk factor at $\mathrm{CR} \leq \sqrt{M}$. The cap equations of Section 5.4 constrain every active value. The deployed market configuration records the current borrow and supply caps.

Insurance credit. The practitioner sets the recognition factor β_j, the floor fraction ϕ_j, the minimum insurer rating, and the aggregate cap on insurance credit. These four quantities are model constants. The recognition factor satisfies $\beta_j \in [0, 1]$. The reduced layer remains at least $\phi_j \lambda_j$, and $\beta_j = 0$ until the practitioner activates the credit.

Appendix B: Symbols

Notation for the methodology. SIMM quantities (WS, K_b, ρ, γ, ψ, CR) follow ISDA SIMM v2.6 [1]. None of these prescribed parameters is fitted.

θ	Liquidation threshold (LLTV); $\theta = \kappa(1 - H)$ ($= 1 - H$ at Prime), with $H_{\text{eff}} = 1 - \theta \geq H$.
$\ell_{\max}$	Maximum borrow LTV; $\ell_{\max} = \theta - \nu$.
ν	Per-listing liquidation buffer between LLTV and LTV (4–8 pp); static rather than dynamic.
Δt_A	Longest gap between usable published valuations in business days: the attested valuation publication frequency (the publication cycle of Section 2.1) plus the attested publication lag. Debt accrues across it every calendar day, so the buffer requirement takes the year fraction $\Delta t_A/252$ (Eq. (11)).
H	Total collateral haircut; $H = \text{IM} + \delta_{\text{clear}} + \delta_{\text{carry}} + \Lambda$.
IM	Delta component of ISDA SIMM v2.6 initial margin per unit of published valuation at SIMM's native horizon of ten business days (no MPOR rescaling), aggregated over risk classes IR, CreditQ, CreditNonQ, Equity, FX and summed over product classes (Eq. (24)).
IM_p	Margin of SIMM product class p, RatesFX and Credit among them: the ψ roll-up over the risk classes it holds. The product-class margins add, $\text{IM} = \sum_p \text{IM}_p$ (Eq. (24)).
δ_{clear}	Clearing-discount component of H, above the liquidation threshold: the difference between the price risk of the governing clearing window and the SIMM charge at ten business days. Day one $\delta_{\text{clear}} = \text{IM}(\sqrt{LH/10} - 1)$, a charge for $LH > 10$ and a margin credit for $LH < 10$, read from contractually enforceable terms, attested or documented. Estimated legs and FRTB fallback windows never earn a credit. Once $\geq N_{\text{obs}}$ qualified clearing observations exist, $\text{IM} + \delta_{\text{clear}} = \max(\text{IM}\sqrt{LH/10},\ \delta_{\text{obs}})$ (Eq. (6)). The discount any clearing party receives is bounded by $\text{IM} + \delta_{\text{clear}} + \delta_{\text{carry}}$, inside H.
δ_{obs}	Conservative quantile of the qualified clearing observations inside the practitioner's lookback, as a total clearing cost per unit of published valuation.
N_{obs}	Number of qualified clearing observations that must exist before δ_{obs} can govern the price-risk charge (Eq. (6)). The practitioner sets it with the quantile, the lookback, and the diversity requirement, as the standard for a qualified clearing observation (free parameter, Appendix A).
δ_{carry}	Interest-cost component of H: the certain cost of holding the debt over the clearing window, $\delta_{\text{carry}} = \theta\, r^{B,\max}\, LH/252$ (Eq. (7)), the ceiling annual percentage yield charged simple over the window's calendar span of $LH \times 365/252$ days, and charged only after liquidation begins. It never enters IM, ν, or λ_{gate}, and is computed in closed form at Eq. (8).
LH	Governing clearing window in business days (Eq. (4)): the sum of the four legs $w + d_n + d_d + d_s$, with the three-source hierarchy and its floors fixed in Section 2.1. The window composes across a fund-of-funds wrapper's and the underlying fund's notice, dealing, and settlement chains at the documented worst case, and terms are never averaged. The window is never shorter than one business day.

Λ	Sum of additive haircut layers; $\Lambda = \lambda_{\text{gate}} + \lambda_{\text{stale}} + \lambda_{\text{oracle}} + \lambda_{\text{def}} + \lambda_{\text{prep}} + \lambda_{\text{ops}} + \lambda_{\text{ww}} + \lambda_{\text{lev}} + \lambda_{\text{event}}$.
λ_{gate}	Gating/suspension layer.
λ_{stale}, $\lambda_{\text{stale}}^{\text{base}}$, $\Delta\lambda_{\text{stale}}$	Valuation staleness/smoothing layer, $\lambda_{\text{stale}} = \lambda_{\text{stale}}^{\text{base}} + \Delta\lambda_{\text{stale}}$: the per-type baseline for the ordinary age of an on-time publication, plus the overdue-publication escalation of Eq. (13), which is zero while publication stays within the contractual publication cycle.
λ_{oracle}	Oracle pipeline integrity layer; set by the oracle-configuration tier, 0 at Robust.
λ_{def}, $\lambda_{\text{def}}^{\text{floor}}$	Default add-on, PD $\times$ LGD over the clearing window LH (private credit only), floored at the practitioner's controlled per-type value $\lambda_{\text{def}}^{\text{floor}}$ (Eq. (17)); zero for the types whose default tail is booked in another layer.
λ_{prep}	Prepayment add-on (amortising mortgage/consumer securitised collateral).
λ_{ops}	Manager/operational layer; its value is a practitioner-set model constant for each type.
λ_{ww}	Wrong-way layer.
λ_{lev}	Leverage add-on (fund-unit collateral only).
λ_{event}	Event-risk add-on (event-linked ILS collateral only): max of the attested 99% annual-aggregate modelled valuation loss (AEP, ultimate, net of fund hedges) and the largest attested modelled single-event valuation impact; floored at a prescribed conservative value when the event-model file is unattested or uncorroborated.
λ_j^{eff}, $R_{\text{ins},j}$	Additive layer λ_j after any qualifying-insurance credit, and the recognised reduction itself: $\lambda_j^{\text{eff}} = \max(\phi_j\lambda_j,\ \lambda_j - R_{\text{ins},j})$ with $R_{\text{ins},j} = \beta_j\,\lambda_j\,\min(1, \text{NetLimit}/E_{\text{ref}})$ (Eq. (30)). Eligible only against the manager/operational layer λ_{ops} and the oracle-integrity layer λ_{oracle}, one policy to one layer (Section 5.3).
β_j, ϕ_j	Prescribed recognition factor for the policy's evidenced quality, $\beta_j \in [0, 1]$, and the prescribed floor fraction below which the credit never takes the layer. Both are practitioner elections, and the credit ships inert at $\beta_j = 0$ until they are set (free parameter, Appendix A).
NetLimit, E_{ref}	The policy's per-occurrence limit *net* of deductibles, retentions, shared-limit allocation, and excluded perils, and the reference exposure it is measured against, $E_{\text{ref}}^{(a)} = \max(\text{live exposure}, S_{\max}^{(a)})$.
r, r_{floor}	Credit grade entering the default add-on, and the prescribed floor grade applied where no admissible rating exists (Eq. (15)). A rating from a nationally recognized statistical rating organization governs where one exists, a corroborated manager internal rating r_{int} resolves one grade weaker and no better than BB+, and the practitioner sets r_{floor} (free parameter, Appendix A).
$\text{PD}_1(r)$	Published one-year cohort default rate at grade r, read from rating-agency default tables and never fitted.
$\text{PD}(r, LH)$	Probability of default over the clearing window, $\text{PD}(r, LH) = 1 - (1 - \text{PD}_1(r))^{LH/252}$ (Eq. (16)): the survival form of a constant default intensity on a 252-business-day year, evaluated only at $LH < 252$.

LGD, $\mathrm{LGD}_{\mathrm{floor}}$ — Loss given default, the fraction of exposure not recovered, read from a published senior-secured recovery table keyed to attested seniority. The prescribed FRTB default-risk-charge values are 75% for senior debt and 100% for non-senior debt or an equity instrument. Where seniority is unattested the claim takes the non-senior value, $\mathrm{LGD}_{\mathrm{floor}} = 100\%$ (free parameter, Appendix A).

R_{cap} — Attested issuer redemption capacity in USD per dealing window, window-normalised where the attestation basis differs, refreshed at the attestation frequency.

g — The offering documents' stated per-cycle gate fraction, used in the pre-signature capacity proxy $g \cdot \mathrm{AUM}$.

$B_{\max}$ — Borrow cap, $B_{\max} \le R_{\mathrm{cap}}$, visible in the on-chain market configuration. Where the deployment can attribute debt to entities, the largest observed single-entity debt is monitored as $B_{(1)} \le R_{\mathrm{cap}}$, a diagnostic rather than an assumption.

$S_{\max}$ — Supply cap; $S_{\max} \le M\, R_{\mathrm{cap}}$.

M — Supply-cap multiple of redemption capacity; bounds the concentration risk factor at $\sqrt{M}$ (free parameter; 8 illustrative).

T — SIMM concentration threshold, re-keyed to redemption capacity: $T = R_{\mathrm{cap}}$.

CR — Concentration risk factor, $\mathrm{CR} = \max\bigl(1, \sqrt{\text{position}/T}\bigr)$.

WS_k — Weighted sensitivity of risk factor k: $WS_k = RW_k\, s_k\, \mathrm{CR}_b$ (published risk weight $\times$ attested sensitivity $\times$ concentration risk factor).

K_b — Within-bucket aggregate margin, $K_b = \sqrt{\bigl(\sum_k WS_k^2 + \sum_{k\ne l} \rho_{kl} f_{kl} WS_k WS_l\bigr)_+}$.

$(x)_+$ — Positive part, $\max(x, 0)$, applied to the sum under the root in K_b (Eq. (21)).

ρ_{kl} — Prescribed within-bucket risk-factor correlation.

f_{kl} — SIMM concentration factor, $f_{kl} = \min(\mathrm{CR}_k, \mathrm{CR}_l)/\max(\mathrm{CR}_k, \mathrm{CR}_l)$.

γ_{bc} — Prescribed across-bucket correlation, applied to bucket sums S_b bounded to $[-K_b, K_b]$.

ψ — Prescribed across-risk-class correlation matrix, applied to the risk classes a SIMM product class holds before the product-class margins add to IM (Eq. (24)).

κ — Due-diligence tier multiplier scaling θ, $\kappa \in (0, 1]$ with the top tier at 1 (free parameter; $\{1.00, 0.90, 0.75\}$ illustrative).

$r^{B,\max}$ — Ceiling of the deployed market's borrow-rate model: its annual percentage yield at full utilisation, on a 365-day calendar basis, at which the position's debt is taken to accrue every calendar day. A kink rate qualifies only where no higher rate is attainable. The deployed market configuration supplies this input between recomputations, and the buffer requirement uses it rather than the live borrow rate r^B (Eq. (11)).

c — Interest cost per unit of debt over the clearing window, $c = r^{B,\max} LH/252$. It enters the closed-form solve $H = (\mathrm{IM} + \delta_{\mathrm{clear}} + \Lambda + \kappa c)/(1 + \kappa c)$ with $\delta_{\mathrm{carry}} = \theta c$ (Eq. (8)).

J, $q_p(J)$	Jump at the next published valuation and the prescribed per-type stressed-jump floor used in the buffer requirement: the lower p-quantile of that jump, set as policy and never fitted to the token's own valuation history (Eq. (11)).
$\hat{\sigma}$	Staleness volatility implied by the type SIMM calibration (Eq. (13)).
ρ_{sub}	Published SIMM sub-curve correlation between distinct sub-curves of one currency ($= 0.993$); enters only where distinct sub-curves of one currency are held, for example a USD Prime sub-curve on securitised collateral.
MPOR	Margin period of risk: the horizon over which margin must cover the collateral's price move.
w, d_n, d_d, d_s	The four legs of LH (Eq. (4)), in business days: the worst-case wait to the next submission deadline, and the attested notice period, dealing window, and settlement lag.
H_0	The haircut before the interest cost, $H_0 = \text{IM} + \delta_{\text{clear}} + \Lambda$, so that $\delta_{\text{carry}} = H - H_0$ exactly (Eq. (8)).
$Q_{\text{cap,red}}$, Q_{issuer}, Q_{queue}	Redemption capacity over one dealing window and its two terms: the issuer capacity term equals R_{cap}, and the committed-settlement term is the within-window processing level the subscription documents commit to, zero where they commit to no measurable level (Eqs. (34) and (35)).
$q_L^{\text{committed}}$	The committed settlement level for one window, read from the subscription documents case by case per listing and recorded as an attested field, or the higher attested commitment where one exists. The feasibility check passes only where it covers the borrow cap.
q_L, q_O^*, A	The redemption request submitted for the position in liquidation, the stressed concurrent third-party demand on the same window, and the fund AUM. The window fills $\min\{q_L + q_O^*,\, g\,A\}$, and q_O^* takes the conservative floor $q_O^* = g\,A$, a fully subscribed window, where the transfer agent does not report queue depth and gate utilisation (Eq. (35)).
$B_{(1)}$, $B_{\text{tot}}^{(a)}$	Largest observed single-entity debt against a collateral, and total outstanding debt against it, $B_{\text{tot}}^{(a)} = \sum_e B_e^{(a)}$.
CovR, u	Coverage ratio $\text{CovR} = R_{\text{cap}} / \sum_e B_e$ and its reciprocal, the utilisation $u = 1/\text{CovR}$: aggregate borrows against one window of redemption capacity.
$S_{\text{global}}^{(\text{issuer})}$	Global supply cap per issuer, bounding the sum of per-deployment supply caps, since R_{cap} is an issuer-level resource that does not multiply with the number of listings (Section 5.4).
NMRF	Non-modellable risk factor, the FRTB treatment of a risk factor without usable data [2]. D-SIMM follows it in the missing-data rule: an unavailable input takes a prescribed conservative floor, never zero (Section 4.6).

References

[1] International Swaps and Derivatives Association, Inc., *ISDA SIMM® – Methodology, version 2.6*, 2023, effective date: December 2, 2023. [Online]. Available: https://www.isda.org/a/b4ugE/ISDA-SIMM_v2.6_PUBLIC.pdf

[2] Basel Committee on Banking Supervision, *Minimum capital requirements for market risk*, 2019, bCBS d457, January 2019. [Online]. Available: https://www.bis.org/bcbs/publ/d457.pdf